

Team Computer Smashers PS SIH26043 Sponsor Government of Jharkhand

Theme Disaster Management Format 24-hour build • 100 marks

Version 4 • final • [decisions log](#)

JharSetu: our playbook for winning SIH26043

Most teams will build a portal where people post problems and others comment on them. We'll build a **challenge exchange: the loop that turns a villager's voice note into a verified, funded, staffed and tracked project**, with a Crisis Mode that switches the whole system to rapid response when an alert comes in.

READ THIS FIRST • ABOUT 10 MINUTES

Start here: everything we're building, in one read

This section is for every member of Computer Smashers. Read it once, all the way through, before the hackathon. By the end you should be able to explain the problem, our solution, why we chose it, every feature and screen, and the plan for the 24 hours, without opening anything else. The rest of the playbook is the detail behind each part.

What we're building

JharSetu is a platform for the Government of Jharkhand, built for problem statement SIH26043. People report local problems by voice note, photo, text, or even a plain SMS when there's no internet. JharSetu turns each one into a clear project, connects it to a university team that can build a fix, a company that can pay for it and an NGO that can deliver it, and tracks the work until the fix is proven on the ground.

It runs in two modes. **Peace Mode** handles long-term problems in education, health, farming and disaster preparedness. **Crisis Mode** switches on when a flood or storm alert arrives, or during a district's mock drill, and turns the platform into a crisis room for urgent needs. It's built for Jharkhand first, but the region is a setting: in our college round we also show it running for Rajkot.

Our goal

THIS WEEKEND

Win our college's 24-hour round, judged out of 100 on the evaluation matrix.

NEXT

Get nominated for SIH 2026 under SIH26043, where the Government of Jharkhand evaluates. Everything we build now carries forward.

WHAT WE DELIVER

A working, deployed prototype of one complete loop, with a real database, real logins and real AI. Not a complete product, and never mock screens.

The problem

Three groups need each other and never meet. **Citizens** see problems first, but their reports go into WhatsApp groups and grievance portals, where they get lost, repeated and never acted on. **Universities** have students and labs, but students build made-up projects because nobody gives them real problems. **Companies** must spend CSR money on social causes, including disaster management, but have no list of verified local problems to fund. And even when someone builds a solution, nothing tracks it to real use, so most projects die after the demo.

The example we use everywhere: lightning killed over 2,400 people in Jharkhand in ten years, most of them farmers. Official alerts exist, but workers in open fields without smartphones never receive them. A university electronics lab could build a cheap village siren that plays those alerts, and a steel company could pay for the sirens. Today none of them know about each other.

Our solution: one loop from report to proven fix



1. **Report.** A citizen sends a voice note, photo or text in Hindi or English, or a plain SMS if there's no internet.
2. **Compile.** AI turns the report into a clear brief and merges duplicates, so 31 reports of the same problem become one challenge.
3. **Verify and prioritise.** Locals and field volunteers confirm it, it gets a priority score with the reasons shown, and a coordinator approves it.
4. **Match.** The platform lists resources already nearby, then recommends a university team, a company and an NGO, with reasons.
5. **Propose and pilot.** Student teams submit proposals, each gets a Readiness Score, and the best one is piloted.
6. **Pledge.** Companies and NGOs each cover part of what's needed until nothing is missing. We call this Resource Swarm.
7. **Prove.** The challenge only closes with photo evidence and a volunteer's check, and then the

community confirms the fix. The result goes into the Impact Ledger, and partners get automatic reports.

Why this solution, and not the obvious one

The obvious idea is a portal where people post problems and others comment or upvote. Most teams will build exactly that, and it fails for five reasons. Each of our choices fixes one of them:

WHY THE OBVIOUS PORTAL FAILS	WHAT WE DO INSTEAD
Raw complaints are too vague for anyone to act on	The Challenge Compiler turns each report into a solvable brief
Upvotes reward loud city areas and bury remote villages	An explainable priority score that weights vulnerability. Popularity counts for only 5%.
Nobody knows who can help, or what help already exists nearby	Resource-first matching, then capability matching with the reasons shown
Ideas pile up and never get built, and nobody checks whether anything was fixed	Readiness-scored proposals, pilots, and closure only with evidence
Universities and companies have no reason to keep using it	Automatic CSR and accreditation reports, credit-bearing projects, a talent pipeline

Three more choices make it fit this problem statement specifically. **Crisis Mode** owns the Disaster Management theme. The **SMS fallback** keeps it working when floods knock out mobile data. And building for **Jharkhand first, with a region switch**, earns the relevance marks while proving any state can use it.

Who uses it

ROLE	WHAT THEY DO
Citizen	Reports a problem in under a minute, follows its status, and confirms when it's fixed
Field volunteer or NGO	Checks reports on the ground with a photo, lists their resources, and helps deliver the fix
Coordinator (district officer or faculty coordinator)	Approves briefs, rates proposals, approves pilots, and closes challenges with evidence
University team	Picks up real problems, forms a team, proposes and pilots a

ROLE	WHAT THEY DO
	solution
Company	Funds or supplies part of a verified project, and gets an impact report
State official	Sees district-by-district analytics of what's reported, who's engaged and what got solved

The winning features

These are what judges will remember. The first three are the ideas we repeat throughout the pitch.

PITCH IDEA 1

Challenge Compiler

A messy Hindi voice note becomes a clear project brief in seconds, showing what the AI is unsure about. A human approves it.

PITCH IDEA 2

Matching and Resource Swarm

It checks what's already nearby first, then recommends partners with reasons. Several partners each pledge part of the need until the gap closes.

PITCH IDEA 3

Impact Ledger

Nothing closes without evidence and community confirmation. Every fix leaves a tamper-evident record, and partners get their reports automatically.

DEMO BEAT

Readiness Score

Two proposals side by side: an impressive AI prediction system scores 48, and a cheap siren relay scores 84. The one that can actually be deployed wins.

FINALE

Crisis Mode drill with SMS

A mock flood alert turns districts red. A report arrives by SMS from a phone with no internet, and lands on the map with an exact location.

EQUITY

Silent zones

The map highlights high-risk areas where almost nobody is reporting, because silence usually means no access, not no problems.

Every feature, by tier

We build strictly in this order. **MUST** features make the loop work. **WOW** extras only start once the whole loop works, in the order listed. **PITCH ONLY** items go on the roadmap slide and are

never built.

TIER	FEATURE	IN ONE LINE
MUST	Six roles and a demo switcher	Switch between citizen, volunteer, coordinator, university, company and admin in one click, with no passwords on stage
MUST	Report a need	Voice, photo, GPS and text, plus people affected, urgency and vulnerable groups. Hindi or English, under a minute, and it still works with no photo or location.
MUST	Challenge Compiler	Speech-to-text, translation and a structured brief, with a rule-based fallback if the AI is down
MUST	Duplicate merging	Reports about the same problem, in the same place, join one challenge
MUST	Confidence level and priority score	How well verified it is, and how urgent it is, with the reasons shown. The coordinator approves in one click.
MUST	Organisation profiles and resource registry	Who can do what, and which boats, drones or supplies are within reach of each challenge
MUST	Matching and team builder	Recommended departments, companies and NGOs with reasons, and a student team with the right mix of skills
MUST	Proposals and Readiness Score	Teams propose, the coordinator rates readiness, and the best proposal gets a pilot
MUST	Resource Swarm	The pilot's needs go out as alerts, partners pledge parts of them, and a live bar shows what's still missing
MUST	Lifecycle and evidence-based closure	A workspace board, milestones and a timeline. A challenge only closes with photos, a beneficiary count and a volunteer's check.
MUST	Impact record	People served, time to match, time to resolution and remaining need, saved when a challenge closes
MUST	Map, list and dashboard	District map coloured by priority, a list view if the map fails, and the headline numbers
MUST	Notifications	Live updates in the app, and status updates to reporters in their own language

TIER	FEATURE	IN ONE LINE
MUST	Seed data and demo reset	Realistic simulated data for both scenarios, one command to reset the demo, and cached AI answers
WOW 1	Crisis Mode	A real alert or mock drill turns districts red and opens the crisis room with needs, gaps and nearby resources
WOW 2	SMS channel	Critical reports go out as a short coded SMS when there's no internet, basic phones can text in, and the platform texts replies and tech-squad call-ups
WOW 3	Region switch	Switch the platform from Jharkhand to a small Rajkot region, with our own university as a partner
WOW 4	Silent zones and hazard layers	Map layers for lightning, flood, drought and mining risk, plus the areas nobody reports from
WOW 5	Community checks and reports	"Still a problem / fixed / wrong" buttons, before-and-after photos, the tamper-evident ledger, and CSR and accreditation PDFs
WOW 6	Pipeline trace panel	Shows each AI step and its timing live, so judges can see the AI working
WOW 7	Offline mode	Reports wait on the phone and send themselves when the internet returns, with a network status bar and a text-only view
WOW 8	Do-not-duplicate library	Similar challenges already solved elsewhere, with their design, cost and lessons
WOW 9	Solution co-pilot	An AI panel that helps student teams with ideas from similar solved challenges. Only if everything above is done.
PITCH ONLY	Roadmap	Tribal languages through Bhashini; missed calls, phone menus, USSD, WhatsApp and Telegram; a native Android app that sends SMS automatically and finds location from the phone tower; a report-helper assistant for citizens; ask-the-data for officials; weather, sensor and satellite data; the state relief portal

Not building at all: a general chatbot that gives advice, disaster prediction, blockchain, native apps in 24 hours, payments, and anything that dispatches rescue (that's 112's job).

The screens

SCREEN	WHAT'S ON IT	MAIN ACTION
Landing	"See a need. Form a team. Close the loop." Four buttons and three live numbers.	Report a need
Report a need	Voice, photo, location, text and checkboxes. A network status bar, and "Send by SMS" for critical reports when there's no internet.	Submit
Challenge detail MAIN DEMO SCREEN	The brief, confidence, priority breakdown, nearby resources, recommended partners, proposals with readiness scores, the gap bar, the timeline and evidence	Whatever the next step is: approve, adopt, pledge or upload
Coordinator dashboard	"Needs immediate action" first, then the ranked queue, the map and list, KPIs, and the region switch	Approve brief
Organisation profile	Expertise, resources on a map, past work, verified badge	Offer resources
Collaboration workspace	A board (Ideas → Under review → Prototype → Field test → Deployed), the team, pledges, files and evidence	Upload evidence
Crisis room	A red "Crisis Mode" band, red districts, needs with their gaps, nearby resources, incoming SMS reports, the tech-squad roster	Accept task or pledge
Impact and analytics	Ledger entries with before-and-after photos, the funnel, the headline KPIs, district and institution breakdowns	Download report

How we stand out

- We turn complaints into completed projects. Most teams stop at collecting them.
- Our AI does specific jobs and shows what it's unsure about. A human always decides.
- We check what's already nearby before anyone builds something new.
- We score proposals for readiness, so the cheap fix that works beats the fancy one that doesn't.
- Nothing closes without evidence and the community's confirmation.
- It works without mobile data, and even on basic phones, through SMS.

- It's built on Jharkhand's real situation (lightning deaths, tribal languages, the state's relief portal), and switches to any region.
- Every partner gets something they already need: CSR reports, accreditation evidence, real projects.

The demo in one minute

1. **Report:** Somra, a farmer in Gumla, records a Hindi voice note about lightning deaths.
2. **Compile:** AI builds the brief live and merges it with 30 matching reports.
3. **Approve:** the coordinator sees why it's critical and approves it in one click.
4. **Match:** siren stock and shelters nearby, then a university lab, a company and an NGO, with reasons.
5. **Propose and pilot:** readiness 48 against 84; the siren relay gets the pilot, and two companies close the gap.
6. **Prove:** before-and-after photos, a volunteer's check, Somra confirms, and the CSR report is generated.
7. **See:** the dashboard with silent zones, then a quick switch to the Rajkot region.
8. **Drill:** a mock flood alert for Sahebganj, an SMS report arriving from a phone with no internet, and the call-up buzzing on stage.

How we'll proceed

WHEN	WHAT HAPPENS	CHECKPOINT
Night before	Accounts and keys, the SMS gateway phone, the seed-data sheet, paper wireframes, everyone reads this playbook	Nothing left to set up on the day
H0 – H1	Lock the scope; nobody writes feature code yet	Everyone knows their next five hours
H1 – H2	Design contract: database schema (with a region field), API routes, a mock service for the frontend	Frontend and backend can work in parallel
H2 – H6	Foundation: database, logins, report screen, map, Compiler fallback, auto-deploy	H6: a report on the deployed site shows up in the feed
H6 – H12	Core loop: compile, dedup, score, match, propose, pledge	H9: the whole path runs with no AI. H12: real AI plugged in.
H12 – H18	Close the loop, then the Wow extras in order	H18: feature freeze

WHEN	WHAT HAPPENS	CHECKPOINT
H18 – H20	Test everything and do the final deploy	H20: every test passes on the live site
H20 – H22	Final deck, backup video, Q&A drill	The story is as finished as the product
H22 – H24	Timed rehearsals, including failure rehearsals; submit; rest the presenter	Ready to present

Your first job, by role

ROLE	OWNS	FIRST TASK
Lead	Scope, story, demo script, research, time	Run the H0 lock meeting and set up the task board
FE	Citizen and coordinator screens, proposals, workspace	App shell and the Report a need screen
UI	Product look, map, dashboards, deck visuals	Theme tokens and the map page
BE	Database, roles, lifecycle, matching, crisis engine	Schema (including the region field) and row-level security
AI	Compiler, dedup, scores, fallback	The rule-based fallback, then the Compiler prompt
INT	SMS gateway, seed data, deployment, QA	Auto-deploy, then the gateway phone

Rules we all follow

- Golden path before extras. Working before pretty. Pretty before new.
- Feature freeze at H18. The Lead has final say on scope.
- Never claim something is built when it isn't.
- Every number on a slide has a source or comes from our own measurement.
- Every task has one owner. If you're stuck for 30 minutes, say so.

Words we use

Challenge

A verified problem the platform tracks. One

Challenge Compiler

The AI step that turns reports into a clear project

challenge can merge many reports.

Confidence level

How well verified a challenge is: unverified, community corroborated, field verified, coordinator approved, resolved with evidence

Coordinator

The district officer or faculty coordinator who approves briefs, pilots and closures

Resource Swarm

Several partners each pledging part of what's needed, with the remaining gap shown live

Pilot

A small real-world trial of the chosen proposal

Crisis Mode, mock drill

The rapid-response mode, triggered by a real alert or a practice drill

Connectivity ladder

How reports get out: internet first, then SMS, then waiting on the phone until a network returns

Golden path

The one demo flow that must always work

Must, Wow, Pitch only

Our build tiers: must work, build if time allows, roadmap slide only

brief

Priority score

A 0–100 urgency score with its reasons shown

Resource registry, "available nearby"

The list of boats, drones, supplies and labs, and which ones are within reach of a challenge

Readiness Score

A 0–100 score for how ready a proposed solution is to deploy

Impact Ledger

The permanent, tamper-evident record of every fix

Silent zone

A high-risk area where almost nobody is reporting

Region switch

Changing the platform from Jharkhand to another region, Rajkot in our demo

Fallback

The rule-based version that keeps running if the AI is unavailable

Guide to this playbook

SECTION	WHAT'S IN IT	READ CLOSELY IF YOU'RE
The winning thesis	Our core argument, the three pillars, the two modes, the story in seven lines	Everyone
Read the problem properly	The sponsor's seven official components and how we cover each, the stakeholders, Jharkhand facts, how we handle the theme, policy links	Lead; everyone before Q&A

SECTION	WHAT'S IN IT	READ CLOSELY IF YOU'RE
The solution	Name, personas, a sample Compiler brief, proposals and readiness, standout features, AI assistants	Everyone
Features, screens, tiers	The full build list with owners and hours, every screen in detail, what not to build	Everyone, especially FE and BE
Product look and feel	Colours, map legend, fonts, interface rules	UI, FE
Demo path and scenarios	The 8-step demo, the lightning and flood scenarios, the Rajkot region, seed data	Lead, FE, INT
Architecture and stack	The diagram, the stack, AI vs code, the connectivity ladder and SMS format, the five formulas, data model, API routes, lifecycle	BE, AI, INT
Security and responsible AI	Who can do what, privacy, AI rules, the audit trail	BE
24-hour battle plan	The shared checklist, phase by phase	Everyone; tick tasks as you finish
Roles and ownership	Who owns, presents and backs up what	Everyone
Scoring playbook	How we earn each of the 100 marks, and the metrics we show	Lead
Pitch and slides	The hook, 5- and 7-minute running orders, slides, comparison, roadmap, sustainability	Lead and presenters
Judge Q&A bank	Likely judge questions with short answers	Everyone
Definition of done	The judge's journey, the submission checklist, the README abstract	Lead, INT
What everyone must know	The knowledge check before judging	Everyone
Risks and team rules	What could go wrong and how we handle it	Everyone

SECTION	WHAT'S IN IT	READ CLOSELY IF YOU'RE
Decisions log	Why we chose what we chose, across all drafts and discussions	Anyone who wants to reopen a decision

If you remember one thing: we're not building a complaint portal. We're building the loop that turns a complaint into a proven fix. Every university becomes the R&D department of its district.

THE WINNING THESIS

A portal won't win this. A working loop will.

Problems don't stay unsolved only because resources are missing. They stay unsolved because the right problem doesn't reach the right people at the right time, and nobody tracks it to completion. Collecting complaints is easy. Turning messy, duplicated, multilingual reports into problems someone can actually solve, and proving the fix happened, is the hard part. If we own that, we win.

REFINE

Challenge Compiler

A voice note, photo, text or plain SMS in Hindi or English is compiled into a structured, deduplicated brief that a student team or a company can act on. It shows a confidence level and what the AI is unsure about, and a human approves it.

MATCH

Capability matching and Resource Swarm

Each challenge is matched to the departments, companies, NGOs and volunteers whose capabilities fit, and to resources already nearby, such as a boat, a drone or siren stock. Partners pledge part of what's needed, and the remaining gap stays visible until it's closed.

DELIVER

Impact Ledger

Teams propose solutions and the readiest one gets piloted. A challenge only closes with evidence and community sign-off, and every closure leaves an auditable record. The CSR and accreditation reports it generates give partners a reason to stay.

Two modes, one platform. In *Peace Mode*, JharSetu runs long-term challenges in education, health, agriculture and resilience. When an IMD flood or thunderstorm warning arrives, or a

district runs a mock drill, *Crisis Mode* turns the affected districts red, opens a crisis room, and calls up university tech squads, nearby resources and industry pledges. This is how we own the Disaster Management theme without drifting away from the problem statement.

The story in seven lines

1. A citizen reports a need.
2. The platform compiles it into a challenge.
3. The community and field volunteers verify it.
4. The system finds the right capabilities, and the resources already nearby.
5. Teams propose solutions, and the readiest is piloted.
6. Several organisations combine resources.
7. The result is tracked until impact is proven.

Pitch discipline. Repeat three ideas all the way through: *Challenge Compiler*, *Capability Matching*, *Impact Ledger*, with Crisis Mode as the finale. Don't claim every feature is new. Claim that the combination is: discovering the problem, collaborating on it, deploying the fix and measuring its impact, all in one loop.

"Every university becomes the R&D department of its district."

That's our closing line. Everyone on the team should be able to say it without looking.

PROBLEM UNDERSTANDING • 10 MARKS

Read the problem properly

SIH26043 comes from the **Government of Jharkhand**. Everything we build, including demo data, languages and examples, has to feel like it was made for Jharkhand's 24 districts. We're a Gujarat team, so this is where most of our research time goes.

What the sponsor explicitly asks for

The problem statement lists seven expected components. Judges will check against this list, so we'll put a slide in the pitch that shows 7 of 7 covered, plus the extras.

OFFICIAL COMPONENT

HOW JHARSETU COVERS IT

SEEN IN
DEMO

OFFICIAL COMPONENT	HOW JHARSETU COVERS IT	SEEN IN DEMO
Citizen engagement module for problem submission (with multimedia evidence)	Voice-first PWA with photo, GPS, text, people affected, urgency and vulnerability; works offline; coded SMS when there's no internet; plain SMS from basic phones; assisted reporting by volunteers	Steps 1 and 8
AI-enabled categorization and routing	Challenge Compiler: transcribe, translate, extract, deduplicate, score, draft the brief, recommend partners and nearby resources	Steps 2–4
University collaboration interface for team formation	Capability matching to departments, a skill-gap team builder, and solution proposals from student teams	Steps 4–5
Industry partnership facilitation	CSR-fit matching; capability-request alerts; partial pledges of money, equipment, people or expertise with a visible resource gap (Resource Swarm); milestone-based release	Step 5
Project lifecycle management	A 9-stage lifecycle: proposals scored for readiness, pilot approval, a shared workspace, evidence-based closure, auto-release of stalled claims	Steps 5–6
Visual analytics dashboard (districts, institutions, outcomes)	District map and list, hazard layers, silent zones, the funnel, an institution leaderboard, a region switch, and two headline KPIs: time to team formed, and share of challenges reaching pilot or deployment	Step 7
Notification and communication system	Reporter updates in their own language, capability-match alerts to partners, SMS replies and tech-squad call-ups, in-app realtime, workspace comments	Throughout

The real problem underneath

The real question is how a district can continuously turn real problems into validated, deployable solutions. Today, three groups need each other and never meet:

COMMUNITIES

They see the problems first: a flooded culvert, a lightning-prone field, a school with no safe drinking water. Their

UNIVERSITIES

They have students, labs and faculty expertise, and NEP 2020 pushes experiential, community-linked learning. But

INDUSTRY

CSR law (Section 135, Schedule VII, which names disaster management explicitly) requires companies to spend, and

reports go to WhatsApp groups and grievance portals, where they stay unstructured, duplicated, unverified and unfunded.

students keep rebuilding library-management projects because nobody hands them real, well-scoped problems.

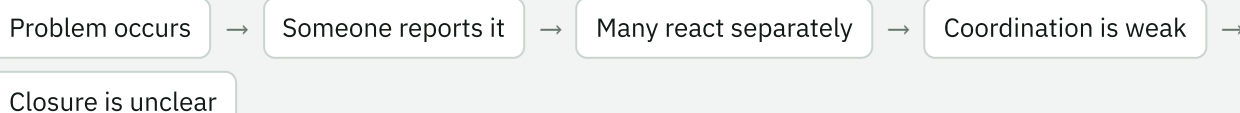
Jharkhand's steel, coal and mining companies have large budgets. What they lack is a supply of validated local problems, capable partners and impact they can report.

Everyone in the loop

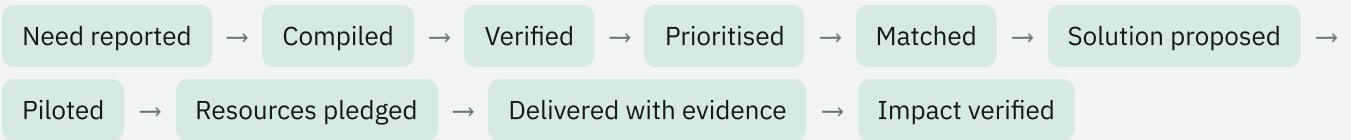
STAKEHOLDER	WHAT GOES WRONG TODAY	WHAT JHARSETU GIVES THEM
Citizens and affected communities	Don't know where to report, or whether anything is happening after they do. Elderly, disabled and remote people are easily overlooked.	Voice reporting in their language, SMS when there's no internet, status updates, and the final say on whether it was fixed
Field volunteers and NGOs (NSS, Aapda Mitra, local NGOs)	Get unstructured, duplicate, incomplete reports, and often duplicate each other's work	Clustered reports, verification tasks near them, a shared workspace
Coordinators (district officers, faculty coordinators)	No live view of unmet needs, who's working on what, or where the resource gaps are	A ranked queue, a map, resource-gap tracking, one-click approval
Universities	Skills and student teams, but no real problems to point them at, and good projects never reach deployment	A filterable challenge board, the team builder, proposals that can become pilots
Industry	Equipment, logistics, experts and CSR funds, but no trusted channel	Validated projects, partial pledges, board-ready impact reports
State departments (JSDMA, Higher and Technical Education)	No picture across districts of what's reported, who's engaged and what got solved, and little documentation to learn from after a disaster	Analytics by district, institution and outcome, and a library of what worked

The loop we're closing

TODAY



WITH JHARSETU



There's a failure here that nearly every team will miss: **solutions die after the demo**. Hackathon prototypes, SIH ones included, rarely reach deployment because nothing tracks them from idea through pilot to verified impact. Closing that loop is the core of our product.

"JharSetu doesn't just collect complaints. It turns them into verified, assignable, measurable challenges that universities and industry can take on."

Jharkhand context we must speak fluently

FACT	WHY IT MATTERS FOR US
Lightning is Jharkhand's biggest natural disaster: over 2,400 deaths between 2014–15 and 2023–24, most of them farmers	Our main demo challenge: getting lightning alerts to farm workers who don't have smartphones, and giving them somewhere to shelter. A good match for ECE and civil departments, and for industry CSR.
In August 2026 the state began piloting a <i>Jharkhand Disaster Relief Management</i> portal in Dhanbad and Bokaro, covering lightning, drowning, snakebite and road accidents	Position JharSetu as a complement: their portal handles relief and compensation, and ours sends the underlying problems to people who can solve them. We'll show it as an integration point.
A large tribal population, with Santali, Mundari, Ho, Kurukh, Nagpuri (Sadri) and Khortha widely spoken, and many basic phones in rural areas	Voice-first reporting, SMS for basic phones, and a language roadmap through Bhashini. Almost no other team will think about this. We'll demo Hindi reliably and show the plan for tribal languages.
Mining hazards: the Jharia coalfield fires, land subsidence around Dhanbad. SIH 2026 itself has a mine-subsidence problem statement (SIH26025).	IIT (ISM) Dhanbad sits next to the problem. This makes a strong example of "the nearest capable university" in our matching demo.
Ganga floods in Sahebganj, drought and heat in Palamu and Garhwa, human–elephant conflict, forest fires	Sahebganj hosts our Crisis Mode drill. The rest go into seed data so the map looks like the real state.
Universities to name as <i>potential</i> partners: IIT (ISM) Dhanbad, BIT Mesra, NIT Jamshedpur,	Realistic matching. Label them "demo data, potential partner", and use fictional company and

FACT

WHY IT MATTERS FOR US

Central University of Jharkhand, Birsa
Agricultural University, Ranchi University, RIMS
Ranchi

NGO names so we never imply a real organisation
pledged anything.

How we handle the theme

Watch the mismatch. Public listings file SIH26043 under both "Disaster Management" and "Smart Education", and the sponsor's background text names education, healthcare and agriculture. If we build only a disaster app, a judge can say we narrowed the problem. The ResQGrid and AegisGrid drafts both frame the product around disasters. That's right for Crisis Mode, but too narrow for the whole product.

Our framing: "JharSetu handles every kind of societal challenge. We launch with *disaster resilience as the flagship vertical*, and Crisis Mode is what no generic platform has." Every challenge gets a resilience tag (the disaster-management phase it belongs to: mitigation, preparedness, response or recovery), and the seed data includes education, health and agriculture challenges too.

Policy alignment for the slides

- **PM's Ten-Point Agenda on Disaster Risk Reduction (2016).** Point 6 is "develop a network of universities to work on disaster-related issues"; others cover using social media and mobile technology, building on local capacity, and learning from every disaster. JharSetu puts several of these into practice. Quote the exact wording from the DST source listed at the end.
- **Sendai Framework 2015–2030.** Understanding risk (the map), governance (the approval workflow), investment (CSR) and "build back better" (the recovery challenges).
- **NEP 2020.** Experiential learning and industry collaboration, cited in the sponsor's own background.
- **Unnat Bharat Abhiyan, NSS and Aapda Mitra.** Existing networks that can do field verification and assisted reporting.
- **CSR Schedule VII (disaster management, relief, rehabilitation)** and the **DPDP Act 2023** for privacy.

INNOVATION • 15 MARKS

The solution: JharSetu

JharSetu joins *Jharkhand* with *setu*, the Hindi word for bridge. Tagline: "**Communities**

raise it. Campuses solve it. Industry scales it."

Lines from the two drafts that stay: from ResQGrid, **"See a need. Form a team. Close the loop."** becomes the landing-page headline and **"From local need to collective action"** becomes the README subtitle. AegisGrid's closing image, the platform as a bridge, is literally what *setu* means, so it becomes part of our close. The decisions log explains why we didn't keep either draft's name.

Other names we considered: *Akhra*, the village gathering ground in Jharkhand's tribal culture (a strong story, but check with someone from Jharkhand that it's appropriate before using it), and *SamadhanSetu*. Lock the name in the first hour and never debate it again.

Personas

REPORTER

Somra, farmer, Gumla district

Speaks Nagpuri and Hindi and owns a basic Android phone. Two neighbours were killed by lightning last monsoon. He will never fill in a form, but he will send a voice note.

FIELD VERIFIER

NSS volunteer or local NGO worker

Lives near the reports and can confirm a need with a photo in five minutes if someone asks. Today nobody does, and she hears about problems days later on WhatsApp.

COORDINATOR

District disaster-management officer, or a faculty coordinator

Has to review hundreds of reports and has no time. Needs clean, deduplicated, ranked problems with their confidence shown, to approve with one click.

SOLVER

Final-year ECE and civil students, plus a faculty mentor

Want real problems for projects, internships and papers, a certificate that means something, credit under NEP 2020, and the chance to see their work deployed.

FUNDER

CSR head at a steel or coal company (fictional in the demo)

Must spend CSR money locally and report on its impact. Wants validated projects, the option to fund part of a solution, milestone control, and a board-ready impact report.

STATE OFFICIAL

JSDMA or Higher and Technical Education department, Ranchi

Needs the picture across districts: what's being reported, which institutions are engaged, and what actually got solved.

What the Challenge Compiler produces

This is the screen judges should remember. Build the challenge detail page to look like this card. The values below are simulated.

PRIORITY 82 · CRITICAL

COORDINATOR APPROVED

AI-ASSISTED

SIMULATED DATA

Last-mile lightning alerts and safe shelter for farm workers, Gumla block

COMPILED FROM	31 reports (voice, photo, text) from 12 villages, corroborated by 2 field volunteers	
PROBLEM	Official lightning alerts exist (IMD, and IITM Pune's Damini app), but workers in open fields without smartphones never get them, and there's nowhere safe to shelter.	
PEOPLE AFFECTED	About 4,800 farm workers and their families	
NEEDS	A siren relay that plays official alerts (electronics) · 12 siren units · a low-cost shelter design (civil) · village training	
AVAILABLE NEARBY	30 siren units in stock at a fictional electrical supplier in Ranchi · community halls that can double as shelters in 4 of the 12 villages	
SUGGESTED PARTNERS	BIT Mesra ECE lab and Birsra Agricultural University extension team (potential partners) · CSR from a fictional steel company · a fictional local NGO	
OUTCOME	Sirens and shelters working in 12 pilot villages before the next monsoon	
SUCCESS METRIC	Alerts reach registered farm workers ahead of storms in every pilot village, and no lightning deaths in pilot villages this season	
RESOURCE GAP	12 siren units: 8 pledged, 4 open	
WHY CRITICAL	12 villages in a lightning hotspot, farm workers with no shelter, and no partner assigned yet	
	Severity 23 /25	Urgency 9 /15
	People 11 /15	Vulnerability 13 /15
	Hazard 10 /10	Resource gap 8 /10
	Recurrence 5 /5	Community 3 /5
AI IS UNSURE ABOUT	The exact number of workers affected (estimated from 31 reports and	

village sizes), and whether two of the villages share a panchayat. A coordinator should confirm both.

AI-assisted recommendation. Final validation is required from an authorised coordinator.

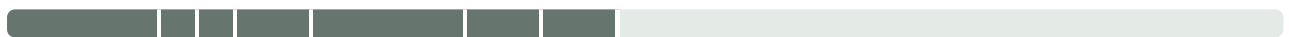
A crisis-time example from the ResQGrid draft, placed in Sahebganj. Input: "Road blocked and medicines can't reach the village." Output: "Set up a temporary last-mile medicine route for 350 residents within 8 hours. Needs: logistics, mapping, local transport. Success: 50 essential medicine kits delivered."

Proposals and the Readiness Score

Once a team is formed, student teams submit proposals, and each one is scored for readiness. A cheap solution that can go live next month beats an impressive one that needs a year. It also quietly answers the teams pitching "AI predicts disasters": we reward what can actually be deployed. Simulated values:

Proposal A • Custom sensor network with AI lightning prediction

48



Technical 12/20

Cost 3/15

Time to deploy 3/15

Local resources 6/15

Safety 12/15

Community 6/10

Scalability 6/10

Proposal B • Siren relay for official IMD/Damini alerts, plus low-cost shelter kits

84

APPROVED FOR PILOT



Technical 20/20

Cost 12/15

Time to deploy 12/15

Local resources 12/15

Safety 12/15

Community 8/10

Scalability 8/10

Standout features

These are what separate us from the "portal plus upvotes" teams. Build the ones in the Must tier, and use the three marked **PITCH** as the ideas we repeat on stage.

FEATURE	WHAT IT DOES	WHY OTHERS WON'T HAVE IT	CRITERION
Challenge	A raw voice note or SMS becomes a	Most teams stop at	Innovation,

FEATURE	WHAT IT DOES	WHY OTHERS WON'T HAVE IT	CRITERION
Compiler PITCH	brief like the card above: title, problem, people affected, needs, outcome, success metric, partners, what the AI is unsure about, plus SDG and Sendai tags.	"AI picks a category"	Tech
Semantic dedup and clustering	47 reports of the same blocked culvert merge into one challenge carrying "47 reporters, 12 photos, 3 villages"	This is real engineering (embeddings plus geo-distance), not a CRUD app	Tech
Confidence ladder and community validation	Every challenge shows how far it's been verified: unverified, community corroborated, field verified, coordinator approved, resolved with evidence. Verified local users can also flag it as still a problem, improved, inaccurate, affecting more people, or needing a solution that won't work here. Low-confidence reports stay visible and labelled.	Honest about what AI can't know. We say "confidence", never "truth".	Responsible Tech
Explainable, vulnerability-aware priority	A 0–100 score with a visible breakdown and a one-line "why critical". It counts children, elderly people, disability, pregnancy, medical dependency, isolation and lack of phone signal. Popularity is capped at 5%.	Others rank by upvotes, which favours loud urban areas	Responsible Tech, Impact
Capability and resource-first matching, with Resource Swarm PITCH	First asks what's already nearby (boats, drones, siren stock, a lab), then who can build what's missing. It recommends a combined team and accepts partial pledges, showing the remaining gap live.	Deployable rather than theoretical. No single organisation has to solve everything.	Innovation, Feasibility
Solution Readiness Score	Scores every proposal on technical feasibility, cost, time to deploy, local resources, safety, community acceptance and scalability, as shown above	Turns "idea collection" into "adoption", which is what the sponsor wants	Feasibility, Impact

FEATURE	WHAT IT DOES	WHY OTHERS WON'T HAVE IT	CRITERION
Impact Ledger PITCH	A challenge only closes with evidence, a beneficiary count and a verifier. Each closure records before-and-after photos, time to match, time to resolution, people served and remaining need. Entries are hash-chained, so tampering shows.	Answers "what happened after the report?", with transparency similar to blockchain but without its build time	Impact, Security
Crisis Mode and mock drills FINALE	A real IMD alert, or a district's mock drill, switches affected districts to a crisis room: needs board with resource gaps, nearby resources, incoming SMS reports, university tech-squad call-up, industry pledges. After the event it drafts preparedness challenges for next season.	Ties the platform to the whole disaster cycle. This is our biggest demo moment.	Innovation, Presentation
Connectivity ladder and SMS	Uses the best connection available: internet, then a coded SMS with exact GPS for critical reports, then waiting on the phone. Basic phones can report by plain SMS.	Most teams assume the internet works during a disaster. It usually doesn't.	Feasibility, UX
Silent Zone detection	Highlights areas with high hazard and high population but almost no reports. That usually means a digital divide rather than no problems, so the platform assigns NSS or Aapda Mitra field surveys there.	A thoughtful equity insight that judges remember	Impact, Responsible Tech
Incentive engine	One click generates a CSR and ESG impact report for companies and extension-activity evidence for university accreditation (NAAC/NIRF)	Answers the question every judge asks: "why would anyone actually use this?"	Feasibility, Sustainability
Region switch	The same platform runs for any region by loading its map, languages and partners. We show Jharkhand and a small Rajkot region.	Proves scalability live instead of claiming it	Scalability

FEATURE	WHAT IT DOES	WHY OTHERS WON'T HAVE IT	CRITERION
Do-not-duplicate library	When a new challenge arrives, shows similar ones already solved elsewhere, with the design, cost, materials, lessons learned and the previous team's contact	Builds institutional memory. It puts Point 9 of the Ten-Point Agenda (learn from every disaster) into practice.	Innovation, Scalability

Demo trick: the pipeline trace panel. When a report is processed, show each AI step as it runs, for example Transcribe 1.9s → Translate 0.6s → Extract 2.1s → Dedup: joined C-112 (similarity 0.91) → Score 82/100 → Draft 3.4s . Judges can't see AI unless you show it, and this panel makes the complexity visible. (Show the real timings from your build, not these.)

AI assistants, not a chatbot

We don't build a general "ask me anything" chatbot. If it made up safety or medical advice during a flood, people could get hurt, and it's the first thing a judge will attack. Instead, the full product has three small assistants, each doing one job:

ASSISTANT	FOR	WHAT IT DOES	SAFETY RULE
Report helper	Citizens	Asks follow-up questions by voice or text to complete a report ("How many people? Since when? Anyone elderly or sick?") and tells them the status	Safety guidance only from official, pre-approved NDMA or IMD cards, shown word for word with the source. It never makes up advice. For anything life-threatening it says "Call 112".
Solution co-pilot	Student teams and mentors	Suggests approaches that worked for similar challenges elsewhere, points to useful datasets, and drafts a proposal and milestone plan	Answers come only from the challenge brief and the do-not-duplicate library, with sources shown. A mentor reviews them.
Ask the data	Coordinators and state officials	Answers questions like "Which blocks in Gumla have critical challenges with no team yet?"	Read-only access to the database, and it shows the numbers it used

In the prototype, none of these is a Must. The solution co-pilot is the last Wow item (about 2

hours), built only if everything else is done. The other two are roadmap. The pitch line: *"We didn't build a chatbot that guesses. Our AI does specific jobs, and a human stays in charge."*

SCOPE DISCIPLINE

Features, screens and build tiers

A reliable working loop scores higher than ten incomplete features. Build strictly tier by tier, and don't start a Wow feature until the whole golden path works end to end. Wow features are listed in the order to build them.

TIER	FEATURE	OWNER	EST. HOURS
MUST	Auth with 6 roles and a one-click demo role switcher (citizen, field volunteer/NGO, coordinator, university, industry, admin)	BE	2
MUST	Report a need: voice, photo, GPS and text, plus people affected, urgency and vulnerability checkboxes. Hindi/English, done in under a minute, and still works with no photo or no location.	FE	3
MUST	Challenge Compiler: speech-to-text, translation, structured brief (category, disaster phase, severity, people, needs, outcome, success metric, uncertainties), with a rule-based fallback that returns the same shape	AI	4
MUST	Deduplication and clustering (embeddings plus distance) with a merged cluster view	AI	3
MUST	Confidence ladder and priority score with breakdown; coordinator queue with one-click approval	AI + FE	3
MUST	Organisation profiles and a resource registry (seeded), with an "available nearby" radius search	UI + BE	2
MUST	Capability matching with reasons, and the skill-gap team builder	BE	2.5
MUST	Solution proposals with a Readiness Score and pilot approval	BE + FE	2.5
MUST	Resource needs from the approved proposal, capability-	BE	2

TIER	FEATURE	OWNER	EST. HOURS
	match alerts, pledges and a live gap bar (Resource Swarm)		
MUST	Lifecycle state machine: workspace board, milestones, evidence-based closure, activity timeline	BE	3
MUST	Impact record on closure (people served, time to match, time to resolution, remaining need)	INT	1.5
MUST	District map with the priority-band legend, a list view fallback, and the KPI dashboard	UI	3
MUST	Notifications: in-app realtime and reporter updates in their language	INT	2
MUST	Seed data for both scenarios, <code>/demo/reset</code> , cached AI responses for demo inputs	INT	2
WOW 1	Crisis Mode, triggered by a real CAP alert or a mock drill: red districts, crisis room with needs, gaps and nearby resources	BE + UI	3
WOW 2	SMS channel: coded SMS fallback for critical reports, plain SMS from basic phones, SMS replies and the tech-squad call-up, all through a gateway phone (see the connectivity ladder)	INT + FE	3
WOW 3	Region switch, and a small Rajkot region for the college round	UI + INT	2
WOW 4	Silent Zone layer and hazard overlays	AI + UI	2
WOW 5	Community validation buttons and sign-off, before-and-after photos, hash-chained ledger, CSR and accreditation PDF reports	INT	3
WOW 6	Pipeline trace panel	FE	1
WOW 7	Offline queue, network status indicator and text-only view	FE	2
WOW 8	Do-not-duplicate library: "similar challenges solved elsewhere"	AI	1.5
WOW 9	Solution co-pilot panel for student teams, grounded in the brief and the library. Only if everything above is done.	AI	2

TIER	FEATURE	OWNER	EST. HOURS
PITCH ONLY	Bhashini for Santali, Mundari and Ho; missed calls, IVR, USSD, WhatsApp and Telegram as extra channels; a native Android app (automatic SMS, tower-based location, phone-to-phone relay); the report helper and ask-the-data assistants; relaying official alerts to registered reporters; weather, sensor and satellite data; inventory systems; volunteer identity verification; DigiLocker certificates; APAAR and Academic Bank of Credits; integration with the state relief portal and national emergency database	Lead (slides)	0

"Pitch only" items go on the roadmap slide and in Q&A answers. Never claim they're built.

Screens to build

SCREEN	WHAT IT SHOWS	OWNER
Landing (public)	"See a need. Form a team. Close the loop." Four actions: Report a need, Explore challenges, Join a solution team, Offer resources. Three live numbers: challenges needing support, organisations contributing, solutions reaching deployment.	UI
Report a need	One screen, or at most three short steps. Record a voice note or type; add a photo; location is filled automatically, with an offline village picker as backup. People affected, urgency, vulnerability checkboxes, consent, and a network status indicator. With no internet, a critical report offers "Send by SMS". No disaster-management jargon.	FE
Challenge detail MAIN DEMO SCREEN	Left: the problem, location, evidence, people affected and the priority explanation. Right: needed skills and resources, what's available nearby, recommended partners, status, and one primary next-action button. Bottom: proposals with readiness scores, discussion, timeline and impact. The pipeline trace opens as a side panel, community validation buttons sit under the evidence, and SMS-sourced reports carry a "via SMS" badge.	FE
Coordinator dashboard	A "Needs immediate action" strip first. Then map with a list toggle; open challenges, critical unmet needs, active responders, resource gaps, time to team formed; filters by district, category, priority band, status and organisation; the region switch in the header.	UI
Organisation profile	Type, location, response radius, expertise, resources with locations	UI

SCREEN	WHAT IT SHOWS	OWNER
	and availability, past challenges completed, verified badge	
Collaboration workspace	A board (Ideas → Under review → Prototype → Field test → Deployed), team members and roles, pledges, comments, files, closure evidence. The solution co-pilot panel lives here if it gets built.	FE
Crisis room	Red band reading "Crisis Mode" (and "mock drill" when it is one), red district map, needs board with resource gaps, nearby resources, incoming SMS reports, tech-squad roster, pledge feed	UI
Impact and analytics (public)	Impact Ledger entries with before-and-after photos, the funnel from reported to verified, the two headline KPIs, and breakdowns by district and institution	UI + INT

What not to build

These are distractions in a 24-hour build. Both drafts listed most of them; the notes explain where we draw the line.

DON'T BUILD	NOTE
Native Android or iOS apps	The PWA installs on phones and is enough. A native app is roadmap, for automatic SMS and tower-based location.
Real satellite image processing, drone integration, full IoT systems, 3D maps	Roadmap slide only. Drones appear as registry resources, not integrations.
Anything that needs government APIs we can't get	We mock the IMD/CAP alert, and label it as a drill
"AI predicts disasters" claims	We coordinate the response; we don't forecast. No validated data, no prediction claims.
Blockchain	The hash-chained ledger gives us tamper evidence without it
Training our own ML models	Use APIs and embeddings; measure accuracy on a small labelled set
A general "ask me anything" chatbot, or AI-written safety advice	See "AI assistants, not a chatbot". Voice <i>reporting</i> in Hindi stays: it's a single input that leads to action, not a conversation.

DON'T BUILD	NOTE
Telegram and WhatsApp bots	Both need internet, which is what fails in a flood. SMS covers every phone. They're roadmap channels.
A dashboard full of made-up metrics, or a fake "live" disaster feed	Every number comes from seeded records that are labelled as simulated
Screens that aren't part of a workflow	If a screen isn't on the golden path or in the Must tier, it doesn't exist
Hand-rolled authentication	Supabase Auth with row-level security
Payments, social feed, hospital management, real-time emergency dispatch	JharSetu never dispatches rescue. That's for 112 and the district administration.
A second full demo scenario	The Crisis Mode finale is a 45-second glimpse of the flood drill, and the Rajkot switch is 10 seconds. Neither is a second walk-through.
A sprawling category list	Eight fixed categories (disaster and safety, water, health, education, agriculture, roads and infrastructure, energy and connectivity, environment), plus hazard tags

UI/UX • 10 MARKS

Product look and feel

The AegisGrid draft set the direction: an emergency operations centre meets human collaboration. The app should feel like a serious, high-trust tool, not a social app. We keep that direction, and change three details so it works for Jharkhand's users.

Ink #102027 Text and icons	Ground #F4F6F5 Background	Line #CCD1C7 Borders, dividers	Teal #2E7180 Brand, actions, coordination
Critical #D94F45 Text version #A8332A	High #E07B2E Text version #9A4A12	Moderate #E5A83B Text version #8A5A00	Long-term #3867A6 Preparedness, recovery

Resolved #3E8064
Text version #2F6B52

Rules

- Red, orange and amber only ever mean urgency. Teal is for actions and coordination. Green only means verified or resolved.
- Map markers are coloured by priority band: critical 75 and above, high 50–74, moderate 25–49, long-term below 25. Green means resolved.
- Colour is never the only signal. Every marker and chip also carries an icon or a word, because red and green look alike to many colour-blind users.
- Use the fill colours for markers and bars, and the darker text versions for words on white. The fill versions don't meet WCAG AA contrast for text.
- Crisis Mode puts a red band across the top ("Crisis Mode · Sahebganj · mock drill"), fills affected districts red and mutes the rest of the map. Peace Mode stays teal.
- Every screen has one obvious next-action button: "Approve brief", "Pledge 4 units", "Upload evidence".
- Critical markers pulse gently, and the pulse stops under reduced-motion settings. No other decorative animation.

Type

Mukta for headings and body. It covers Latin and Devanagari in seven weights, so Hindi and English look like one design. Add **Noto Sans Ol Chiki** when Santali arrives. **IBM Plex Mono** for coordinates, IDs, timings and status codes. On citizen screens, use at least 16px body text, 1.6 line height for Devanagari, and tap targets of at least 48px.

What we changed from the draft's theme

- **Ground:** a neutral off-white (#F4F6F5) instead of warm cream (#F4F1EA). It reads as an operations tool rather than a magazine, and warm cream is one of the most common template looks right now.
- **Contrast:** added darker text versions of red, amber and green. The draft's red (#D94F45) is only about 4.1:1 on white, below the 4.5:1 that WCAG AA requires for normal text.
- **Type:** Mukta instead of an editorial display face, because the app has to render Hindi cleanly on every screen.
- **Legend:** the draft used blue for "recovery or preparedness", which mixed urgency with phase. Now blue just means the long-term (lowest) priority band, and each colour means one thing.

Demo path and seeded scenarios

This is the one flow that must work every time. Build it first, rehearse it most, and seed the data so it tells a story. It uses quick role switches, not every feature. The full version runs 4 minutes 35 seconds; the 5-minute pitch uses a shorter cut (see Pitch). Start straight on the report screen, not the landing page.

STEP	WHAT JUDGES SEE	WHAT WE SAY (ONE LINE)	TIME
1 · Report	On a real phone mirrored to the projector, Somra records a Hindi voice note ("lightning killed two people in our field, there's no shelter"), ticks "farm workers, elderly" and attaches a photo with GPS. Optionally, turn on airplane mode, submit, then reconnect so it syncs.	"No form. No English. It even works without a network."	0:30
2 · Compile	The pipeline trace runs live. The report joins the cluster "Lightning risk to farm workers, Gumla", which now has 31 reporters, and its confidence moves up to <i>community corroborated</i> . The brief shows what the AI is unsure about.	"31 voices, one problem, no duplicates."	0:30
3 · Approve	Switch to the coordinator. The ranked queue shows the priority breakdown and the "why critical" line. The brief is approved in one click.	"The score isn't a black box. AI prepares it, a human decides."	0:30
4 · Match	"Available nearby" shows siren stock in Ranchi and community halls that can serve as shelters. Matching recommends a BIT Mesra ECE lab (demo data), a fictional NGO and a fictional steel company's CSR, each with reasons. The team builder forms a team of 2 ECE, 1 CSE, 1 civil and a mentor.	"We don't just ask who can build something. We ask what's already nearby."	0:30
5 · Propose and pilot	Two proposals sit side by side: a sensor network with AI prediction (readiness 48) and a siren relay with shelter kits (84). The coordinator approves the pilot for the second, and its needs go out as capability requests. Switch to industry: they pledge 8 of the 12 siren units, leaving a gap	"The cheap solution that works next month beats the clever one that needs a year."	0:45

STEP	WHAT JUDGES SEE	WHAT WE SAY (ONE LINE)	TIME
	of 4. A second fictional company pledges 4, and the gap closes.		
6 · Prove	Jump to a seeded pilot that's further along, with before-and-after photos. Evidence is in, a field volunteer confirms, and the coordinator marks it deployed. Somra gets a Hindi notification and confirms the fix. The Impact Ledger entry appears, and one click produces the CSR PDF.	"It can't be closed with a button. It closes with evidence."	0:35
7 · See	The district dashboard: the two headline KPIs, the conversion funnel, and silent zones pulsing in two blocks. Then switch the region to Rajkot: the map moves, and local challenges appear with our own university as a partner.	"Silence isn't safety. And the same platform runs for any state, including this one."	0:30
8 · Drill	Finale: click "Run mock drill: IMD heavy-rain alert, Sahebganj". The red band appears, the districts turn red, and the crisis room opens with the Scenario B needs and their gaps. A teammate with mobile data and Wi-Fi off sends a critical report: it arrives by SMS and lands with an exact GPS pin and a "via SMS" badge. A university environmental lab accepts water testing, and the tech-squad call-up SMS buzzes a phone on stage.	"This is the difference between collecting a complaint and coordinating a solution."	0:45

Scenario A: Peace Mode (the main loop)

Lightning safety for farm workers, Gumla (simulated). There are 31 reports across 12 villages in Hindi and English, with deliberate near-duplicates so clustering has something to do. The compiled brief and the two proposals are the ones in the Solution section. Also seed one pilot for this challenge that's already at the evidence stage, with before-and-after photos, so step 6 doesn't have to wait.

Scenario B: Crisis Mode drill (the finale)

Simulated Ganga flood, Sahebganj, adapted from both drafts. The river has overflowed and cut off three villages, about 1,250 people. The main bridge is damaged and mobile data is down in two areas. The crisis room opens with four needs:

- **Drinking water, Village A.** About 200 people with no safe water for two days, including elderly people and children. The main road is partly blocked.
- **Medicine route.** 350 residents cut off, including an elderly patient who needs regular medication. The compiled brief is the "last-mile medicine route" example in the Solution section.
- **Connectivity.** No mobile data at two locations, which is why reports from there arrive by SMS.
- **Shelter.** Temporary shelter capacity is short.

Seed two SMS reports into the drill: a coded SMS with exact GPS from a phone with no data, and a plain Hindi SMS from a basic phone that names only the village, so it starts at a lower confidence. The live SMS in step 8 is a third.

The "available nearby" panel lists 2 boats with a fictional NGO 6 km away, a university robotics lab's drone team that can map flooded roads, and a volunteer with a satellite phone. The drinking-water need is the one that shows Resource Swarm:

PART OF THE SOLUTION	WHO	STATUS
Water-quality testing and a distribution plan	University environmental science lab (potential partner)	Accepted
10 filtration units	Fictional water-equipment company	6 pledged · 4 open
Last-mile distribution by boat	Fictional local NGO	Confirmed

The university's role here is testing, mapping and planning, not delivering relief. Universities don't deliver relief in 12 hours, and a judge will notice if we pretend they do.

From crisis to preparedness: when the drill closes, the platform drafts a follow-up challenge, "a drone medicine corridor for Sahebganj's flood-isolated villages before next monsoon", with safe landing points and OTP handover. That's the AegisGrid scenario, placed where it belongs: as a planned project with time to get drone flight approvals, not something improvised mid-flood.

Also seed one Impact Ledger entry from an earlier simulated flood, so the ledger isn't empty: 200 people served, 42 minutes to match, 9 hours to resolution, 3 photos plus a field confirmation, and a follow-up to check the water supply again in 48 hours.

The Rajkot region (for the college round)

Our first round is judged at our college in Rajkot, so we also seed a small second region. It gives

local judges something they relate to straight away, and it proves the platform isn't tied to one state.

- **Seed:** the Rajkot district boundary (it's in the same map file), about 15 reports, our own university, one fictional company and one fictional NGO, and one fully compiled challenge.
- **Local challenges:** monsoon waterlogging in low-lying wards; heat stress for construction and delivery workers; summer water shortages; fire-safety checks in crowded public places. Frame fire safety as prevention, and don't use the 2024 game-zone fire as a demo prop. Label everything as simulated.
- **On stage:** 10 seconds at the end of step 7 in the 7-minute cut. In the 5-minute cut, keep it as a live answer if a judge asks.
- **If a judge asks "Why Jharkhand, when you're based here?":** "The sponsor is the Government of Jharkhand, so we built for their districts. But the platform isn't tied to one state. Here it is running for Rajkot." Then switch regions.

Background seed data

About 150 more reports across Jharkhand districts, including education, health and agriculture challenges, so the platform clearly isn't disaster-only. Add 10 universities, 8 fictional companies, 5 fictional NGOs and volunteer groups, 40 registry resources (boats, drones, generators, water sensors, 3D printers, satellite phones, siren stock), and 20 solved challenges for the do-not-duplicate library. Every screen that shows seeded data carries a small "Simulated scenario" label.

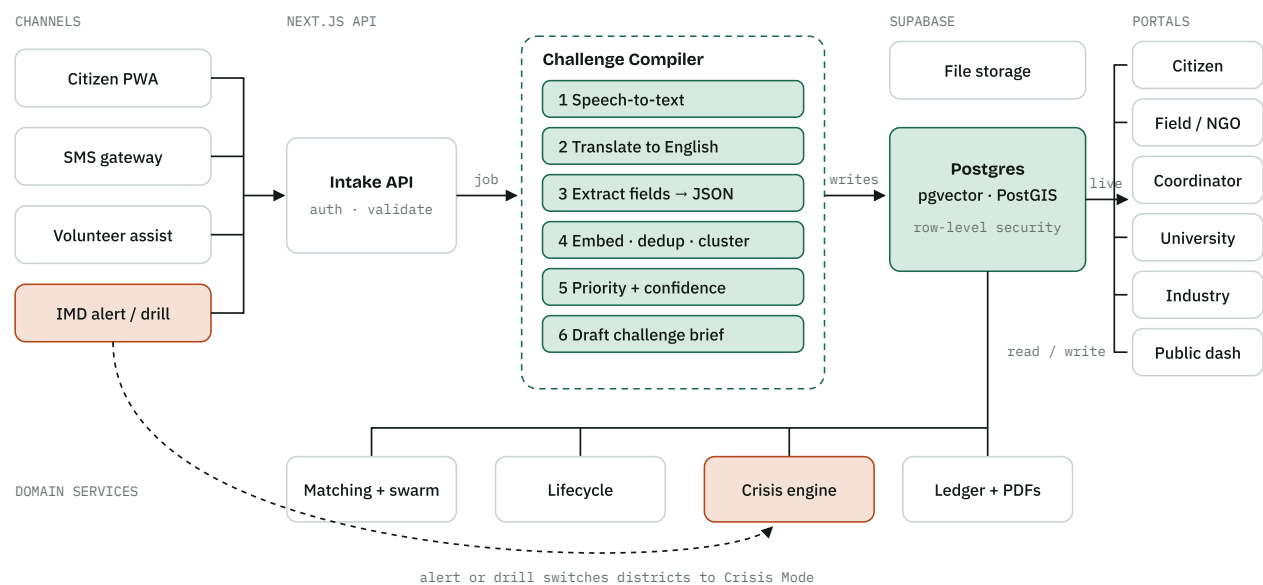
Non-negotiables. A `/demo/reset` script that restores the seeded state in one command. A demo role switcher so nobody types passwords on stage. AI responses cached for the exact demo inputs, so a slow API can't kill the demo. A backup SMS report already sitting in the crisis room, in case the live SMS is slow. A screen recording of the full path as backup. A phone hotspot as backup Wi-Fi. At least two teammates who can run the demo alone. Rehearse three failure versions too: slow network, AI down, and a judge interrupting.

TECHNICAL IMPLEMENTATION • 25 MARKS

Architecture and stack

Choose tools for speed, and avoid depending on too many external services. Supabase gives us Postgres, auth, role-based security, file storage, realtime, vector search and PostGIS out of the box, which saves about six hours of backend plumbing. And because it's open source, it can be self-hosted on government cloud, which is a good answer when

judges ask about data sovereignty.



Every channel goes through one intake API, so adding IVR, missed calls or WhatsApp later means writing a new adapter, not a rewrite. The Compiler runs as a queued job and stays off the request path. Portals subscribe to Postgres changes, so the coordinator's queue, the gap bars and the map update live on stage. PostGIS answers "what's within 20 km of this challenge?"

Stack

LAYER	PICK	WHY, AND THE FALLBACK
Frontend	Next.js (App Router, which is React) + TypeScript + Tailwind + shadcn/ui, installable as a PWA	One repo for the UI and API routes, and it deploys to Vercel in minutes. Plain React with Vite and Supabase also works if the team prefers it.
Backend and data	Supabase: Postgres, Auth, row-level security, Storage, Realtime, pgvector, PostGIS	Realtime replaces Socket.IO, and Auth with row-level security replaces hand-rolled JWT roles. If the team is stronger in MERN, use Node/Express with MongoDB Atlas (vector and geo search). Use the stack you know; it beats a shiny one you don't.
LLM	Claude Haiku 4.5 for fast extraction and Claude Sonnet 5 for drafting, with structured JSON through tool use. Use the Gemini free tier if API credits are a problem.	Keep every LLM call behind one <code>llm.ts</code> function so the provider can be switched in five minutes. There's always a deterministic fallback (below).
Speech-to-	Whisper large-v3 via Groq (fast, has	Roadmap: Bhashini or Sarvam for Indian and

LAYER	PICK	WHY, AND THE FALLBACK
text	a free tier)	tribal languages. Record the demo voice notes clearly in Hindi.
Embeddings	Translate to English first, then embed and store in pgvector	Using one language makes deduplication reliable. Keep the original text for display.
Geo	PostGIS, enabled as a Supabase extension	Radius queries for "available nearby", silent-zone grids and public location fuzzing, all in SQL
Maps and charts	Leaflet + OpenStreetMap + leaflet.heat, district GeoJSON for Jharkhand and Rajkot (datameet), Recharts	Free, no API key. If tiles fail, the list view and a static district map still work.
SMS, PDF, offline	A spare Android phone running an SMS-gateway app (prototype); @react-pdf/renderer; a service worker with an IndexedDB outbox	SMS works on every phone, with or without data. In production, SMS moves to a licensed provider with TRAI DLT registration. Telegram and WhatsApp need internet, so they're roadmap.

AI suggests, code decides

AI DOES (SUGGESTIONS A HUMAN CAN EDIT)	DETERMINISTIC CODE DOES (DECISIONS)
Extracts location, category, people affected and vulnerability, including from free-text SMS	Authentication and authorisation
Summarises reports, writes the challenge brief and says what it's unsure about	Priority calculation. The formula is code; AI only proposes the severity input, and the coordinator can override it.
Suggests the capabilities needed	Lifecycle status transitions
Translates	The evidence required before closure
Finds similar reports and similar solved challenges	Audit log and ledger
Supplies the capability-fit input for matching, and pre-fills cost and time from proposal text	Readiness score from human ratings, parsing the coded SMS, final closure and impact numbers

Build the flow on the fallback first. By H9 the whole golden path runs end to end on rule-based fallbacks, with no AI calls at all: a keyword classifier over the eight categories, rule-based capability mapping, tag matching and cached outputs. Real AI then plugs in behind the same functions by H12. That way the fallback is guaranteed to exist, and AI trouble can never block the demo. If a judge asks, switch the AI off live.

Working without internet: the connectivity ladder

In floods, mobile data usually fails first, while calls and SMS keep working: SMS needs very little bandwidth, and the network holds each message until it can deliver it. So JharSetu always uses the best connection the phone has.

WHAT THE PERSON HAS	WHAT JHARSETU DOES	WHAT GETS THROUGH	IN THE PROTOTYPE?
Internet, even slow	Normal upload: text and location first, then photo and voice	Everything	Yes
No internet, but phone signal	For critical reports, the app writes a short coded SMS with the GPS location and opens the SMS app ready to send. One tap.	Category, severity, exact location, people affected, vulnerable groups, a few words	Yes
A basic phone without the app	A plain SMS in Hindi or any language to the same number. The Compiler reads it.	The text. Location comes from the village name or a registered number.	Yes
Can't read or type	A free missed call from a registered number gets a call back, or a toll-free voice menu in Hindi	"Village X needs contact"	Roadmap
No signal at all	The report waits on the phone. It sends itself when the internet returns; if only phone signal returns, the app offers to send critical reports by SMS.	Everything, later	Yes

How the offline app works

1. On the first online visit, a service worker saves the app, the report screen, translations and the district's village list on the phone.
2. A submitted report goes into the phone's local storage (IndexedDB) with a unique client ID, marked "waiting to send".

3. The app retries whenever a connection appears. On Android Chrome, background sync sends it even after the app is closed.
4. Text and location upload before the photo and voice, so the important part arrives first.
5. The server ignores repeats of the same client ID, so a report retried five times appears once.

Limit: the app has to be opened once online to save itself. NSS and Aapda Mitra volunteers would install it for villagers during pre-monsoon drives.

The coded SMS

Only critical reports use SMS (severity 4–5, or vulnerable groups ticked). SMS can't carry photos or voice, it costs the sender a little, and coordinators shouldn't be flooded. Everything else waits for the internet. The coded message fits in one SMS (160 characters):

```
JS1 K7F2 W5 25.2481,87.6412 P200 VCE "pani nahi 2 din"
```

JS1	JharSetu message format, version 1
K7F2	report ID, so the full version merges with it later
W5	category W (water) at severity 5
25.2481,87.6412	GPS location, accurate to about 10 m
P200	200 people affected
VCE	vulnerable groups: C children, E elderly (also D disability, P pregnancy, M medical dependency)
"..."	a few words ("no water for 2 days")

The server replies: *"JharSetu: report K7F2 received. For life-threatening emergencies, call 112."*

When the internet returns, the full report with photos syncs and merges with the SMS version through the shared ID, so the problem isn't counted twice.

What GPS can and can't do

GPS is one-way: satellites transmit and the phone only listens. The same is true of India's NavIC. So GPS can't send a report, but it works without internet (it's just slower to find a first position), and it gives the exact location that goes inside the SMS. When GPS can't get a fix, a native Android app could use the ID of the tower the phone is connected to, looked up in a tower database like OpenCellID, for a rough location within a few kilometres. A website can't read tower information or send an SMS without the user tapping Send, so both of those are native-app roadmap. In the prototype, the backup is an offline village picker.

Satellite messaging from ordinary phones isn't generally available in India, and personal satellite devices are restricted. Licensed satellite phones sit with district control rooms and response teams.

Prototype setup (about 3 hours, INT)

- A spare Android phone with a SIM, on Wi-Fi, runs an SMS-gateway app (for example the open-

source SMS Gateway for Android) that forwards every incoming SMS to `/api/sms/inbound`. Its number is "the JharSetu number" for the demo.

- The server parses the coded format, and sends any other text to the Challenge Compiler as free text, starting at a lower confidence.
- The PWA shows "No internet. This is critical: send by SMS?" and opens the SMS app pre-filled through an `sms:` link. The link syntax differs slightly between Android and iPhone, so test both.
- The same phone sends the SMS replies and the tech-squad call-up in Crisis Mode.

Limits to admit if asked: SMS isn't guaranteed when the network is badly jammed, only much more likely to get through than data. At real scale in India, SMS needs a licensed provider with TRAI DLT registration; the gateway phone is only for the prototype.

The five formulas everyone must be able to explain

PRIORITY (0-100), each factor normalised to 0-1, computed in code

25 × severity	(AI proposes 1-5, coordinator can override) 5 immediate threat to life · 4 serious harm to health, safety or livelihood · 3 essential service down (water, school, road) · 2 inconvenience · 1 minor
+ 15 × urgency	(deadline, worsening fast, Crisis Mode boost)
+ 15 × people affected	(log scale)
+ 15 × vulnerability	(children, elderly, disability, pregnancy, medical dependency, isolated / no transport, no phone signal)
+ 10 × hazard exposure	(from the GIS layer)
+ 10 × resource gap	(no alternative support yet: share of need unpledged)
+ 5 × recurrence	(seasonal or repeated)
+ 5 × community signal	(unique reporters, log scale, CAPPED)

CONFIDENCE (separate from priority: "is it real?", not "how urgent?")

unverified → community corroborated (≥ 3 independent reporters in the cluster)
→ field verified (volunteer or NGO photo) → coordinator approved
→ resolved with evidence. An "inaccurate" flag from verified locals drops a level until someone checks. Plain SMS from unknown numbers starts at the bottom. Low confidence is labelled, never hidden.

DEDUP same cluster if cosine ≥ 0.85 AND distance ≤ 2 km AND ≤ 30 days
0.75-0.85 means "possible duplicate", decided by the coordinator

MATCH (0-100), the coordinator can override

40 × capability fit	(embedding similarity: needs vs profile)
+ 20 × availability	(capacity now, semester load, stock)
+ 15 × proximity	(distance decay, within response radius)
+ 15 × resource fit	(how much of the open gap they can cover)
+ 10 × partner-type bonus	university: track record + declared interest company: CSR focus + operating district NGO / volunteers: local presence + response time

Resource-first: registry items within the radius are listed before anyone proposes building something new.

READINESS (0-100) per proposal, each factor rated 1-5 by the coordinator or a mentor; cost, time and local resources are pre-filled from the proposal

- 20 × technical feasibility
- + 15 × cost feasibility
- + 15 × time to deploy
- + 15 × local resource availability (from the registry)
- + 15 × safety
- + 10 × community acceptance (from community validation)
- + 10 × scalability

Let the admin edit the weights, and log every change. That's an easy governance and responsible-AI point to make.

Core data model

regions	id, name, center, zoom, boundary geojson, languages[]
users	id, role, org_id, region_id, language, district, skills[], reputation
organizations	id, region_id, type(univ company ngo govt volunteers), name, district, lat, lng, response_radius_km, csr_focus[], csr_budget, verified, embedding
org_capabilities	org_id, capability, capacity, available_from
resources	id, org_id, type, quantity, geom, availability
reports	id, client_id, region_id, reporter_id?, channel(web sms volunteer), phone_hash?, original_text, lang, audio_url, photo_urls[], geom, people_est, urgency, vulnerable[], translated_text, extracted jsonb, embedding, cluster_id, consent, created_at
challenges	id, region_id, title, brief jsonb, category, dm_phase, district, geom, people_est, severity, priority, score_breakdown jsonb, confidence, status, report_count, capabilities[], mode, crisis_id?
verifications	challenge_id, by_user, kind(field still_exists improved inaccurate more_affected unsuitable), evidence_url, note, at
matches	challenge_id, org_id, score, reasons jsonb, status
assignments	challenge_id, org_id, role, accepted_at (the team)
solutions	id, challenge_id, team_id, title, approach, cost_estimate, deploy_days, risks, ratings jsonb, readiness, status
resource_needs	id, challenge_id, solution_id?, item, qty_needed, unit
pledges	need_id, org_id, qty, kind(money equipment people expertise), status
milestones	challenge_id, solution_id, title, due, status, evidence_url
evidence_files	challenge_id, uploaded_by, url, phase(before after closure), at
impact_records	challenge_id, people_served, time_to_match_min, time_to_resolution_min, remaining_need, verified_by
ledger	id, entity, entity_id, action, actor, payload jsonb, prev_hash, hash, at (audit log + timeline, hash-chained)
crisis_events	id, region_id, hazard, source, is_drill, districts[], severity, started, ended
notifications	user_id, channel(app sms), template, payload, sent_at

One region has many organisations, reports and challenges. One crisis event has many reports. Reports cluster into one challenge. A challenge has many matches, assignments, proposals, needs, pledges, milestones and ledger entries, and exactly one impact record once it closes. Status changes and updates are written as ledger rows, and the timeline UI reads from there, so there's no separate updates table. Add `region_id` in the first two hours; adding it later is painful.

API routes

POST /api/reports	submit from the app (idempotent on client_id)
POST /api/sms/inbound	gateway phone forwards every incoming SMS here
GET /api/reports/:id/trace	pipeline steps and timings
GET /api/regions	regions for the switch
GET /api/challenges	filter by region, district, category, band, status,
GET /api/challenges/:id	
POST /api/challenges/:id/approve	coordinator approves the brief
POST /api/challenges/:id/validate	field verification or community signal
GET /api/challenges/:id/nearby	registry resources within the radius
GET /api/challenges/:id/matches	recommendations with reasons
POST /api/challenges/:id/adopt	an organisation joins the team
POST /api/challenges/:id/solutions	a team submits a proposal
POST /api/solutions/:id/review	readiness ratings
POST /api/solutions/:id/approve-pilot	creates the resource needs, alerts partners
POST /api/challenges/:id/pledges	partial pledge against a need
POST /api/challenges/:id/updates	progress note or milestone change
POST /api/challenges/:id/evidence	upload before / after / closure evidence
POST /api/challenges/:id/deploy	coordinator closes it (evidence required)
POST /api/challenges/:id/confirm	community sign-off
GET /api/challenges/:id/report.pdf	CSR / accreditation report
GET /api/organizations/:id	
GET /api/dashboard/metrics	
POST /api/crisis/start	real alert webhook, or { drill: true }
POST /api/demo/reset	

Lifecycle

REPORTED → REFINED → VERIFIED → OPEN → TEAM_FORMED → SOLUTION_PROPOSED → PILOT → DEPLOYED → IMPACT_VERIFIED

- **REFINED** means the Compiler has drafted the brief. **VERIFIED** means the coordinator has approved it, so the AI draft always comes before the human check.
- **SOLUTION_PROPOSED**: teams submit proposals, a coordinator or mentor rates their readiness, and one is approved for pilot. **PILOT**: the prototype is built and field-tested against milestones.
- **DEPLOYED** needs a completion note, a photo or document, a beneficiary count, and

confirmation from a field volunteer or the coordinator. **IMPACT_VERIFIED** is the community's sign-off.

- **Fast path for crisis needs:** OPEN → TEAM_FORMED → DEPLOYED. Delivering water can't wait for proposals, but the evidence rule still applies.
- Side paths: DUPLICATE (merged into a cluster); NEEDS_FOLLOW_UP (deployed, but a follow-up task opens, such as rechecking water supply in 48 hours); CLOSED_NOT_ACTIONABLE (only a human can set this, with a logged reason, and the reporter is told why). AI never rejects a report.
- A claim with no milestone progress in 14 days is released automatically, so nobody can claim a challenge and sit on it.

SECURITY, PRIVACY AND RESPONSIBLE TECHNOLOGY · 5 MARKS

Security and responsible AI

It's only 5 marks, but it's where judges decide whether to trust everything else. About an hour of work and one slide.

Who can do what

ROLE (DEMO ACCOUNT)	CAN	CAN ' T
Citizen	Create reports, see their own submissions and status, flag or confirm a fix	See other people's details or exact locations
Field volunteer / NGO	Verify reports nearby, add evidence, take field tasks, list resources	Approve briefs or close challenges
Coordinator	Approve briefs, set confidence, rate readiness, approve pilots, override priority and matches (logged), close with evidence, start a drill	Edit or delete ledger history
University / industry	Browse open challenges, adopt, propose, pledge, post updates on challenges they're assigned to	See reporters' identities or phone numbers; update challenges they aren't on
Admin	Verify organisation accounts, manage regions, edit formula weights (logged)	Delete ledger entries

Enforce these rules with Postgres row-level security, not just by hiding buttons. Then a judge who asks "what if someone calls the API directly?" gets a real answer. Organisations can't adopt

or pledge until an admin has verified the account.

Privacy

- Never shown publicly: phone numbers, exact locations of vulnerable people, medical details, identity documents, images without consent.
- The public map fuzzes locations to about 500 m. Exact coordinates are visible only to assigned responders and coordinators.
- Public copies of photos have their GPS metadata stripped. The original is kept privately, only for verification.
- SMS senders' numbers are stored hashed for rate limiting and repeat detection, and shown only to coordinators when they need to call back. SMS from unknown numbers starts unverified.
- A consent step before any media upload, anonymous reporting as an option, and only the fields we need, following DPDP Act 2023 principles. Uploads are checked for file type and size.

Responsible AI

- Every AI recommendation opens an explanation: why it's rated this way, which data was used, what the AI is unsure about, who should verify it, and the suggested next action.
- Every AI output carries the same label: "AI-assisted recommendation. Final validation is required from an authorised coordinator."
- AI never dispatches resources and never rejects a report.
- Citizens never receive AI-written safety or medical advice. Only official NDMA or IMD guidance, shown word for word with its source, and "call 112" for anything life-threatening.
- The priority weights are public, popularity is capped, and every weight change is audited.
- Every report and comment has a "report abuse" option, and SMS is rate-limited per number.

Auditability

The hash-chained ledger records who created a report, who changed its priority, who verified it, who rated and approved a proposal, who accepted the task, who uploaded evidence and who closed it.

Live proof on stage: switch to the university role and show that the reporter's phone number is hidden and the location is approximate. It takes ten seconds and makes the security slide believable.

24-hour battle plan

There's a checkpoint every few hours. If one slips, cut scope from the Wow tier, never from the golden path. The app is deployed from hour 6 onwards, so deployment can't surprise us at the end. Tick tasks off as you finish them.



● Ticks are saved on this device.

Before the clock starts

night before

Goal: nothing but ideas and accounts are left to set up on the day. · 0 of 6 done

- ☐ Read the hackathon rules on pre-written code. Accounts, API keys and research are always allowed.

Lead
- ☐ Create Supabase, Vercel, Groq and LLM API accounts, and test each key once. Get a spare Android phone with an SMS-capable SIM, install the SMS-gateway app, and check it forwards a test SMS.

INT
- ☐ Build a seed-data sheet: Scenario A (31 Gumla lightning reports and two proposals), Scenario B (Sahebganj flood needs and two SMS reports), 150 background reports, 15 Rajkot reports, 10 universities, 8 fictional companies, 5 fictional NGOs, 40 registry resources, 20 solved challenges

INT · AI
- ☐ Download the district GeoJSON (Jharkhand and Rajkot) and sketch approximate hazard zones for lightning, drought, flood and mining

UI
- ☐ Wireframe the 8 screens on paper, starting with the challenge detail page

UI · FE
- ☐ Skeleton slide deck and a first draft of the demo script. Everyone reads this playbook, starting with "Start here".

Lead

Kickoff and lock

H0 - H1

Checkpoint H1: nobody writes feature code until the MVP is agreed, and nobody spends more than this hour on alternative ideas. · 0 of 4 done

- ☐ Lock the name, the golden path, both scenarios and the tier list. After this, the Lead has final say on scope.

A11
- ☐ Confirm the stack as written in this playbook, or switch now, and never again

A11

- ☐ Set up the repo with a protected main branch and feature branches. Share `.env` privately and never commit it. INT
- ☐ Set up the task board (To do · In progress · Blocked · Done). Every task gets exactly one owner. Stand-ups every 3 hours. Lead

Design contract

H1 - H2

Checkpoint H2: frontend and backend can work in parallel without waiting on each other. · 0 of 5 done

- ☐ Write the DB schema (with `region_id` and the report `client_id` from the start) and the API contract into the README, and add a typed mock service so frontend builds against it from minute one BE · AI
- ☐ Finalise the component list and wireframes for the 8 screens UI · FE
- ☐ Set up the product theme tokens (colours, Mukta, Plex Mono), the wordmark and base shadcn/ui components UI
- ☐ Freeze the seed-data format (CSV columns match the tables) and the coded SMS format INT
- ☐ Agree the minimum happy path: Report → Compile → Verify → Match → Propose → Pilot → Pledge → Deploy → Impact All

Foundation

H2 - H6

Checkpoint H6: a report submitted on the *deployed* site is stored and appears in the challenge feed. · 0 of 8 done

- ☐ Tables (with PostGIS and pgvector), row-level security policies, seed script, auth, and the demo role switcher BE
- ☐ The rule-based fallback first, then a Compiler prompt that returns the same strict JSON. Test both on 20 real-style reports in Hindi and English. AI
- ☐ Speech-to-text route: upload audio, get text and language back AI
- ☐ App shell with routing, the Report a need screen (voice, photo, GPS, vulnerability) and the challenge feed FE
- ☐ Map page with district boundaries, seeded clusters and the priority-band legend, plus the list view UI
- ☐ Gateway phone connected: every incoming SMS reaches `/api/sms/inbound` and is INT

stored as a report

☐ Deploy the skeleton to Vercel, with every merge to main deploying automatically

INT

☐ Research facts for the slides and lock the problem slide

Lead

Core loop

H6 - H12

Checkpoint H9: the whole golden path runs end to end on the rule-based fallback, with no AI calls. **Checkpoint H12:** real AI is plugged in, and steps 1–5 run in a clean incognito browser. It can be ugly. Show this to mentors. · 0 of 10 done

☐ Embeddings, dedup and cluster merge; priority and confidence with a breakdown JSON

AI

☐ Brief drafter with uncertainties; matching scorer with reasons; skill-gap team builder

AI · BE

☐ Resource registry and the "available nearby" radius query

BE

☐ Solution proposals, readiness ratings and pilot approval

BE · FE

☐ Lifecycle state-machine endpoints (including the crisis fast path), assignments and milestones

BE

☐ Resource needs from the approved proposal, capability-match alerts, pledges and the live gap bar

BE · FE

☐ Coordinator queue, challenge detail page, matching screen with reasons

FE

☐ Heat layer, KPI tiles, conversion funnel, organisation profile page

UI

☐ Coded SMS parser, plain SMS into the Compiler, and the SMS acknowledgement reply

INT

☐ Label 40 test reports by hand and measure category accuracy and dedup precision. These real numbers go on a slide.

AI · Lead

Close the loop, then Wow

H12 - H18

Checkpoint H18: feature freeze. The first item finishes the Must tier, and the rest are Wow features in priority order. After H18, nothing new goes in; only fixes and polish. Nap in pairs for 90 minutes each, and keep at least four people awake. · 0 of 11 done

☐ Evidence-based closure, impact record, activity timeline from the ledger (Must)

BE · INT

☐ Crisis Mode: real alert or mock drill, red band and districts, crisis room with

BE · UI

Scenario B needs, nearby resources and incoming SMS reports, SMS call-up

- ☐ "Send by SMS" in the PWA when offline: builds the coded SMS with GPS and opens the SMS app. Test on Android and iPhone. FE
- ☐ Region switch and the small Rajkot region UI · INT
- ☐ Silent Zone layer (expected vs actual reports) and hazard overlays AI · UI
- ☐ Community validation buttons, sign-off, before-and-after photos, hash chain, CSR and accreditation PDFs INT
- ☐ Pipeline trace panel, offline submit queue, network status indicator, text-only view FE
- ☐ Hindi and English UI strings on all citizen screens FE · UI
- ☐ Do-not-duplicate panel on the challenge page AI
- ☐ Solution co-pilot panel, only if everything above is done AI
- ☐ Architecture slide, comparison slide and first full pitch run-through Lead · UI

Test and final deploy

H18 – H20

Checkpoint H20: every item below passes on the production URL. · 0 of 12 done

- ☐ A new report submits on the web and appears in the coordinator's queue INT
- ☐ A coded SMS from a phone with data off, and a plain Hindi SMS from a basic phone, both appear within seconds, and the sender gets the reply INT
- ☐ With the AI key removed, the fallback still produces a usable brief AI
- ☐ A report with no photo, and one with location permission denied, both still submit FE
- ☐ Recommendations and nearby resources appear with reasons; proposals, pilot approval, pledges and the gap bar all work BE
- ☐ Status changes show in the timeline; evidence upload and deploy work; the impact record appears BE
- ☐ Restricted actions fail for the wrong role, including direct API calls BE
- ☐ The dashboard, map and region switch update live; the mobile layout works on two real phones UI
- ☐ No API keys in the repo or the browser bundle INT
- ☐ Loading, empty and error states on every golden-path screen; consistent UI · FE

spacing and terms

- ☐ Security pass: zod validation, upload type and size limits, rate limiting (including per SMS number), photo GPS stripping BE
- ☐ Final production deploy; `/demo/reset` tested; AI responses cached for demo inputs; backup SMS report seeded INT

Deck, backup and Q&A

H20 - H22

Goal: the story is as finished as the product. · 0 of 4 done

- ☐ Final deck: 7-of-7 slide, architecture, measured accuracy, security, KPIs, three-phase roadmap Lead · UI
- ☐ Record the full golden path as a backup video, and take screenshots FE
- ☐ README: abstract, architecture image, schema, API list, setup steps, demo credentials, team roles AI
- ☐ Q&A drill: the Lead fires the question bank at each person All

Rehearse and submit

H22 - H24

Goal: timed run-throughs on the deployed site, every member speaks, everything submitted, presenter rested. The final hour is for reliability, not new ideas. · 0 of 6 done

- ☐ Three timed rehearsals on the deployed site, finishing 30 seconds under the limit All
- ☐ Failure rehearsals: one on a slow network, one with AI switched off, one with a teammate interrupting as a judge All
- ☐ Charge laptops and phones (including the gateway phone), set up the hotspot, run the demo reset, zoom the browser to 125% INT
- ☐ Final push, tag `v1.0-demo`, and make sure every member has commits in the history Lead
- ☐ Work through the submission checklist in "Definition of done" Lead
- ☐ The presenter rests for 30 minutes before judging Lead

Roles and ownership

These roles assume six members, the SIH team size. Everyone owns one module and presents it themselves, and the commit history should show that. Every task on the board has exactly one owner.

ROLE	OWNS	PRESENTS	BACKUP FOR
Lead · product and pitch	Scope decisions and the feature-creep veto, story, demo script, research, stand-ups, the board, time, resolving disagreements. Also writes some code; the Lead shouldn't be purely a manager.	Hook, problem, close	Demo driver
FE · frontend lead	Report a need (including "Send by SMS"), challenge detail, coordinator queue, proposals, collaboration workspace, trace panel, offline queue	Live demo driver	UI
UI · UX, maps, dashboards, deck design	Product theme, landing, dashboard, map layers, region switch, organisation profile, crisis room, accessibility, Hindi/English strings, plus deck visuals, the architecture diagram and screenshots	Dashboard and silent zones	FE
BE · backend lead	Schema, row-level security, auth, lifecycle, matching, registry, proposal and pledge APIs, crisis engine, ledger writes	Architecture and security	INT
AI · Compiler and matching	Speech-to-text, Compiler, fallback rules, embeddings, dedup, priority and confidence, similar-challenge search, accuracy numbers, the solution co-pilot if time allows	AI pipeline and accuracy	BE
INT · integrations, data and QA	SMS gateway and parser, notifications, PDFs, hash chain, deployment, seed data (including Rajkot), QA checklist, demo credentials, reset, backup video	Connectivity ladder, impact, scalability, roadmap	AI

With five members, merge INT's QA and data duties into the Lead and its integrations into BE. With four, also merge UI into FE. Every area keeps an owner either way.

Scoring playbook

Each criterion in the evaluation matrix needs visible proof in front of the judges. If they can't see something, it doesn't count.

CRITERION	MAX	OUR MOVES	PROOF ON STAGE
Problem understanding and relevance	10	The difference between reporting a problem and solving it; why universities and industry are needed; Jharkhand-specific framing; one clear scenario rather than many vague ones; in scope vs roadmap	Slides 2–3: the lightning figures, the 7-of-7 table, Somra, and "doesn't just collect complaints"
Innovation and creativity	15	Three repeated ideas (Compiler, Matching with Resource Swarm, Impact Ledger), plus readiness scoring, the SMS fallback and Crisis Mode as the finale. The differentiation is the combination of discovery, collaboration, deployment and impact.	Comparison slide; 48 vs 84 readiness; the gap bar closing live; the drill finale with a live SMS
Technical implementation	25	Real AI with a working fallback, real database, deployed URL, queued pipeline, state machine, role-based permissions, geo queries, an SMS gateway, evidence tracking, audit history, typed code, clean README. No screen that isn't wired to a real workflow.	Pipeline trace with real timings, the architecture diagram, our own accuracy numbers, the SMS landing with an exact pin, AI switched off in Q&A if asked
UI/UX and usability	10	The first screen explains the product; one next action per screen; a report takes under a minute with almost no typing; basic phones can report by SMS; critical items are obvious; Hindi/English; readable contrast; colour never alone	The citizen flow on a real phone; the coordinator approves in one click
Feasibility, scalability and impact	15	Free-tier stack; works without mobile data; the region switch proving any state can run it; the three-phase roadmap; the sustainability model; the KPIs below	The live switch to Rajkot; roadmap and KPI slide; the CSR PDF in hand
Security, privacy	5	Everything in the Security section	One slide, plus the live role

CRITERION	MAX	OUR MOVES	PROOF ON STAGE
and responsible technology			switch showing hidden phone numbers and fuzzed locations
Presentation, demonstration and communication	15	Story-led pitch, not a feature list; a rehearsed demo that skips the landing page; backup video; failure rehearsals; Q&A drill; 30 seconds under time	Everything above, delivered calmly
Teamwork and hackathon execution	5	Clear roles, every member presents, a visible board, commits from everyone, integration discipline, testing and deployment	Team slide with each person's module; show the commit graph if asked

Easiest marks to pick up: security and teamwork (10 marks combined) cost about an hour of effort and one slide each. Most teams skip them, so we won't.

Impact metrics we track and show

Headline KPIs: (1) time from report to team formed, and (2) share of challenges that reach pilot or deployment. Put these two first on the dashboard and on slide 11, with the number of university–industry collaborations formed next to them, since that's literally what the problem statement asks for.

SPEED

Time from report to brief ·
brief to match · acceptance
to resolution

QUALITY

Share of reports verified ·
duplicates merged ·
category accuracy on our
labelled set

OUTCOME

People served · vulnerable
people supported ·
resource gaps closed ·
unmet challenges
remaining

In the demo these come from seeded, simulated data. Say so, and present them as what the pilot will measure. We deliberately don't show "estimated cost saved", because we couldn't measure it honestly.

Pitch and slides

We don't know the time limit yet, so there are two cuts: 5 minutes and 7 minutes. Always cut from slides, never from the demo. Tell a story, not a feature list.

Opening hook (Lead)

"Last monsoon, lightning killed two farmers in Somra's village in Gumla. He posted about it in his WhatsApp group. Forty kilometres away, BIT Mesra's electronics students were building a final-year project nobody would use. And a steel company in Jamshedpur had CSR money earmarked for disaster management with no project to put it in. These three never met. JharSetu is where they meet."

Second beat, from the AegisGrid draft: *"The problem is often not a lack of ideas or resources. It's that the right problem doesn't reach the right people at the right time."*

Somra is a representative persona, not a real person. Present him that way, and use the real lightning statistics on the next slide.

Running order

SEGMENT	5-MIN CUT	7-MIN CUT	SPEAKER
Hook, the problem, and why Jharkhand	0:40	1:00	Lead
Solution in one line: Report → Compile → Verify → Match → Pilot → Deploy → Measure	0:20	0:25	Lead
Live demo. The 5-minute cut runs steps 1, 2, 3, 5, 6 and 8 at a brisk pace. The 7-minute cut runs all eight, including the Rajkot switch in step 7.	2:45	4:35	FE drives, UI and AI narrate their parts
Architecture, AI suggests and code decides, resource-first matching, the connectivity ladder, accuracy numbers	0:30	0:30	BE, AI, INT
Impact, roadmap and sustainability	0:25	0:20	INT
Close	0:20	0:10	Lead

Close (from the AegisGrid draft, adapted): *"JharSetu isn't another reporting app. Setu means bridge: between the people who see a problem, the people who can solve it, and the resources to make it real. Every university becomes the R&D department of its district."* In the 7-minute cut, keep only the last sentence.

Slide list

SLIDES 1-4

1. Title, tagline and team. 2. The problem, told through Somra. 3. Jharkhand in numbers, and the loop that's broken today. 4. Solution: Challenge Compiler, Capability Matching with Resource Swarm, Impact Ledger, plus Peace and Crisis modes.

SLIDES 5-8

5. Coverage of the 7 official components. 6. Standout features. 7. Architecture diagram and the connectivity ladder. 8. The AI pipeline, the formulas and our measured accuracy.

SLIDES 9-12

9. Security and responsible AI. 10. Comparison with existing platforms. 11. Headline KPIs, the three-phase roadmap and sustainability. 12. Team roles, then "Thank you. Johar!" (the traditional Jharkhand greeting).

RULE

No slide carries more than about 25 words. Diagrams and screenshots do the talking. Keep backup slides after the "Thank you" for likely questions: costs, IP policy, the data model, SMS limits, drone rules.

The comparison slide

CAPABILITY	GRIEVANCE PORTALS	IDEA-CONTEST PLATFORMS	SIH ITSELF	TYPICAL TEAM'S PORTAL	JHARSETU
Problems come from citizens	Yes	No	No (from ministries)	Yes	Yes, including voice and SMS
Works without mobile data, even on basic phones	Rarely	No	No	No	Yes, through SMS
Turned into solvable problem statements	No	Partly	Yes, by hand	No	Yes, by AI with human approval
Routed to	No	No	Once a year	Browse	Continuous,

CAPABILITY	GRIEVANCE PORTALS	IDEA - CONTEST PLATFORMS	SIH ITSELF	TYPICAL TEAM'S PORTAL	JHARSETU
universities with skill matching				only	with reasons shown
Checks what's already nearby before building new	No	No	No	No	Yes, resource registry
Proposals scored for readiness before a pilot	No	Judged once	Judged once	No	Yes, with pilot approval
Several partners each covering part of a need	No	No	No	No	Yes, with a live resource gap
Tracked to deployment and verified impact	Closure only	No	No	No	Yes, evidence-based closure
Crisis mode and silent-zone equity	No	No	No	No	Yes

Roadmap (slide 11)

PHASE 1 · PILOT

One Jharkhand district for one monsoon season, with one or two Jharkhand universities, a local NGO and one CSR partner

PHASE 2 · STATE

JSDMA and the Higher and Technical Education department; verified industry partners; Santali, Mundari and Ho through Bhashini; SMS at scale through a licensed provider; missed calls, IVR and USSD

PHASE 3 · INTEGRATIONS

A native Android app (automatic SMS, tower-based location, phone-to-phone relay); IMD and state alert feeds; weather, sensor and satellite data; the state relief portal; volunteer identity verification; white-label for other states

Sustainability model

- **Public good first:** free for citizens, owned by the state as shared digital infrastructure rather

than rented.

- **CSR-funded challenge programmes and industry-sponsored challenges** pay for the projects themselves. CSR rules also allow part of a company's CSR spend to go on administrative overheads, which can cover platform and monitoring costs. Check the current cap before stating a number.
- **Grants** from disaster-risk-reduction funds and programmes.
- **White-label licences** for other states and large NGOs (the region switch is the proof), and **premium analytics** for industry ESG dashboards and university accreditation packs.
- **Talent pipeline:** companies recruit students who solved real problems for them. That value keeps them coming back.

Q&A IS PART OF THE 15 PRESENTATION MARKS

Judge Q&A bank

Answer in two sentences, then stop. The person who owns the module answers its questions.

How is this different from a grievance portal, or from SIH itself?

A grievance portal collects complaints and forwards them, and SIH sources problems from ministries once a year. We source problems from citizens continuously, turn them into solvable challenges, pilot the readiest solutions, and track them all the way to verified impact.

How is this different from an emergency helpline like 112?

A helpline sends rescue for a single call. JharSetu handles problems that need knowledge, equipment or funding over hours to months, and routes them to universities and companies. The two work together: a problem behind many helpline calls can become a challenge afterwards.

Why use AI at all?

It turns voice notes, photos, texts and SMS into a usable challenge in seconds instead of days of manual reading, and it powers duplicate detection and matching. It never makes the decision; an authorised coordinator does.

Does it have a chatbot?

Not a general one, because a chatbot that invents safety advice during a flood is dangerous. The full product has three focused assistants: a report helper for citizens

that only shows official guidance, a solution co-pilot for student teams, and ask-the-data for officials.

Why would universities and companies actually use it?

Because it produces what they already have to produce: accreditation and extension-activity evidence, credit-bearing student projects, and CSR and ESG impact reports. Companies also get local projects they can fund in part, and a pipeline of proven student talent.

How do you stop fake or spam reports?

Every challenge shows a confidence level, from unverified up to field verified and coordinator approved, and verified locals can flag a report as inaccurate. Deduplication, photo metadata checks, rate limits per account and per SMS number, and reporter reputation all feed into it, and nothing reaches partners without human approval.

What if the AI gets it wrong?

AI only suggests. Every brief shows what the AI is unsure about, a human approves it, the original reports are linked, and anything can be edited. On our own test set we measured [X]% category accuracy. Fill in the real number.

What happens if the AI service goes down?

The platform falls back to a keyword classifier, rule-based capability mapping and tag matching. Priority, workflow and closure are plain code, so they never depended on AI in the first place. We can switch it off right now and show you.

What happens without internet?

The app works offline and sends reports when the internet returns. If there's no internet but the phone still has signal, critical reports go out as a short coded SMS with the exact GPS location, and people with basic phones can report by plain SMS.

Can GPS send the report when there's no network?

No. GPS only receives signals from satellites; it can't transmit. We use it to put the exact location inside the SMS, which is how a one-line text drops a pin within about 10 metres.

Isn't SMS unreliable in disasters?

It isn't guaranteed, but it needs very little bandwidth and the network holds it until it can deliver, so it gets through far more often than data. After the 2010 Haiti

	earthquake, survivors texted their needs to a free SMS number, 4636, and volunteers mapped them.
Why SMS, and not WhatsApp or Telegram?	WhatsApp and Telegram need internet, which is usually the first thing to fail in a flood, and many villagers use basic phones. SMS works on every phone. Other channels plug into the same intake API later.
What about people without smartphones?	They can report by plain SMS in Hindi, and NSS, Aapda Mitra and local volunteers can report on their behalf. Silent Zone detection tells us where to send people to collect reports in person.
Doesn't Crisis Mode overlap with official disaster response?	No. We never dispatch rescue. We send technical, knowledge and resource needs to universities and industry, under the district authority's direction. It complements the state relief portal rather than replacing it.
Is the drill button just a demo trick?	No. Districts run mock drills before the monsoon, and drill mode lets them rehearse with universities, companies and NGOs on the same platform, clearly labelled as a drill.
Why not deliver medicines by drone during floods?	Drones appear in our crisis room as a nearby resource for mapping flooded roads. Drone deliveries need aviation approvals for flights beyond line of sight, so we turn a drone medicine corridor into a planned preparedness challenge instead of improvising mid-flood.
How can the government use this?	District officers get the ranked queue, the crisis room and mock drills. JSDMA and the higher-education department get analytics by district, institution and outcome. It also gives the state relief portal somewhere to send problems that need a solution, not just compensation.
How does matching work? Isn't it just keyword search?	First it lists resources already within the response radius. Then a weighted score ranks partners: 40% capability fit from embeddings, plus availability, proximity, how much of the gap they can cover, and a partner-type factor. Reasons are shown for every match,

and the coordinator can override it.

What happens after a team submits a solution?

It's scored for readiness, reviewed by a coordinator or mentor, piloted against milestones, and only marked deployed with evidence and community sign-off. The goal is adoption, not a pile of ideas.

Isn't the priority score biased?

The weights are public, popularity counts for only 5%, vulnerability counts for 15%, and every weight change is audited. That's specifically so remote tribal areas aren't drowned out by louder urban areas.

How will impact be measured?

Every closed challenge has an Impact Ledger record: time to match, time to resolution, people served and remaining need, confirmed by evidence and by the community. The two headline KPIs are time from report to team formed, and the share of challenges that reach pilot or deployment.

How does this scale across the state and beyond?

Regions are a setting: a new state loads its map, languages and partners, and we can switch to Rajkot right now to show it. The API servers are stateless, the AI runs as queued jobs, Postgres handles millions of rows, and it can be self-hosted on government cloud.

Why Jharkhand, when you're based in Rajkot?

The sponsor is the Government of Jharkhand, so we built for their districts. But the platform isn't tied to one state. Here it is running for Rajkot. (Switch regions live.)

Can this go beyond disasters?

It already does: education, health and agriculture challenges run through the same loop. Disaster resilience is the flagship because that's where speed matters most.

Who pays for it?

The state runs it as public infrastructure. CSR-funded challenge programmes and sponsored challenges fund the projects, grants cover the rollout, and other states and NGOs can license a white-label version.

How do you protect sensitive location data?

Public maps show locations fuzzed to about 500 m, public photos have their GPS data stripped, and SMS numbers are never shown publicly. Exact coordinates

are visible only to assigned responders and coordinators, enforced in the database with row-level security.

Who owns the intellectual property in the solutions?

By default, solutions are open-licensed as public goods. Where a university or company needs to protect IP, they agree terms when the match is accepted.

What stops a team from claiming a challenge and abandoning it?

Claims are released automatically after 14 days with no milestone progress, reputation drops, and industry funding is only released when milestones are met.

What happens after the hackathon?

We'd pilot it in one Jharkhand district for a monsoon season, with one or two Jharkhand universities, a local NGO and one CSR partner, and measure time to team formed, how many challenges reach deployment, and people served.

FINISH LINE

Definition of done

We're ready when a judge could run this whole journey on the deployed site without a developer's help. Tick it off during the H18–H20 test window.

The judge's journey

on the production URL

Every step works in a clean browser. • 0 of 11 done

- ☐ Log in as a citizen and submit a need with a photo FE
- ☐ See the compiled challenge, its confidence level and what the AI is unsure about AI
- ☐ Log in as the coordinator, check the priority breakdown and approve FE
- ☐ See nearby resources and recommended partners, with their reasons BE
- ☐ Log in as a university team, submit a proposal, and see its readiness score; the coordinator approves the pilot BE
- ☐ Log in as a company, receive the capability alert, pledge against a need, and watch the gap shrink BE
- ☐ Update progress and upload before-and-after evidence FE

- ☐ The coordinator marks it deployed, and the citizen confirms the fix BE
- ☐ See the impact record and download the report INT
- ☐ Turn off mobile data, send a critical report by SMS, and see it land on the crisis map with an exact pin INT
- ☐ Switch the region to Rajkot and back UI

Submission checklist

before H24

All of it, with SIH26043 and "JharSetu" spelled the same everywhere. · 0 of 8 done

- ☐ Deployed app link, demo credentials for all six roles, and the JharSetu SMS number INT
- ☐ Repository link, with a README containing the abstract, setup steps, schema and API list AI
- ☐ Architecture diagram and security explanation BE
- ☐ Seeded data labelled as simulated INT
- ☐ No API keys exposed; environment variables set on the host INT
- ☐ A backup local version that runs without the internet INT
- ☐ Slide deck and backup demo video Lead
- ☐ A summary of each member's contribution Lead

README abstract

Subtitle: *From local need to collective action.*

JharSetu is an AI-assisted, low-bandwidth platform for the Government of Jharkhand. It turns citizen and field reports into verified, prioritised, solvable challenges, and routes them to universities, industry, NGOs and volunteers. A Challenge Compiler converts voice, photo, text and SMS reports in Hindi or English into structured briefs, and a connectivity ladder keeps reporting working without mobile data: critical reports go out as a coded SMS with exact GPS, and basic phones can report by plain SMS. Resource-first matching lists what's already nearby, then recommends partners by skills, availability, location and resource fit. Student teams submit proposals, each scored for readiness, and the readiest is piloted. Resource Swarm lets several organisations each pledge part of a solution while the remaining gap stays visible, and an Impact Ledger records evidence-based closure and

community sign-off. Disaster resilience is the flagship use case: Crisis Mode switches affected districts to rapid response when an alert arrives or a district runs a drill. Regions are configurable, so any state can run it. Unlike grievance portals and reporting apps, JharSetu closes the loop from reported need to verified impact.

SHARED BRAIN

What everyone must know

Any judge can question any member. Every one of us should be able to answer these without hesitating. If you've read "Start here", you already know most of it.

THE PRODUCT

The one-liner, the story in seven lines, the three pitch ideas, the two modes, the closing line, and the seven official components along with where each appears in the demo.

DISASTER MANAGEMENT BASICS

The four phases: mitigation, preparedness, response, recovery. The authority chain: NDMA at national level, JSDMA for the state, then district authorities. The Sendai Framework's four priorities, and point 6 of the Ten-Point Agenda. Why mock drills matter.

ECOSYSTEM

CSR under Section 135 and Schedule VII. NEP 2020's push for experiential learning. Unnat Bharat Abhiyan, NSS and Aapda Mitra. The DPDP Act 2023 in one sentence.

JHARKHAND

24 districts. Lightning is the top killer, with over 2,400 deaths in a decade. The state relief portal pilot in Dhanbad and Bokaro. Ganga floods in Sahebganj. Tribal languages and basic phones. The Jharia fires and IIT (ISM) Dhanbad.

THE SYSTEM

The architecture in 30 seconds. AI suggests, code decides. The connectivity ladder, the coded SMS, and why GPS can't send anything. The five formulas, and why confidence is separate from priority. The lifecycle, its crisis fast path, and the evidence rule. The two headline KPIs.

THE DEMO

At least two people can run the whole golden path alone, in both the 5- and 7-minute cuts, including the live SMS and the region switch. Everyone knows the reset command, the demo credentials and where the backup video is.

Risks and team rules

RISK	MITIGATION
AI or speech API is slow, rate-limited or out of credits on stage	The flow was built on the fallback first; cached responses for demo inputs; a second provider ready behind <code>11m.ts</code>
The live SMS arrives late on stage	A backup SMS report is already in the crisis room, and the backup video shows the live version
The gateway phone dies or loses Wi-Fi	Fully charged and on the venue Wi-Fi with a hotspot as backup; a second phone with the gateway app installed
Map tiles fail to load	List view and a static district map
Wi-Fi drops	Phone hotspot, a local build that runs offline, and the backup video
Hindi speech-to-text makes mistakes	Record the demo voice notes clearly in advance and practise them. Show the transcript being edited, which makes the human-in-the-loop point for us.
A photo or location is missing	The form still submits; the report just starts at a lower confidence
A judge interrupts mid-demo	Rehearsed. The demo reset can jump to any step.
Database integration is late	Frontend builds against the typed mock service from H2, so the UI never waits
Demo data looks fake	Label it as a simulated scenario and make it realistic: real districts, plausible numbers, real hazard types
Two people build the same thing	Contracts at H2, one owner per task, and the board
Scope creep at 3 a.m.	Feature freeze at H18, and the Lead has final say on scope. New ideas go on the roadmap slide.
Merge conflicts break main	Small PRs, merge every 2–3 hours, one person owns integration, and nobody edits another person's files without asking
Last-minute bugs	Deployed since H6, final deploy by H20, freeze at H18

RISK	MITIGATION
Exhaustion	Nap in pairs for 90 minutes, eat on schedule, keep water on the desk, and rest the presenter before judging

Team rules

- Five-minute stand-up every three hours: what's done, what's next, what's blocking you.
- If you're stuck for more than 30 minutes, say so out loud. Nobody fights a bug alone.
- Golden path before Wow features. Working before pretty. Pretty before new.
- Never claim something is built when it isn't. Judges respect "that's on our roadmap", and they penalise being caught out.
- Every number on a slide needs a source or our own measurement.

VERSIONS 2 TO 4

Decisions log

We merged two AI-written drafts, ResQGrid (version 2) and AegisGrid (version 3), and then made our own calls in team discussion (version 4). These tables show where ideas disagreed and why each call went the way it did, so nobody has to re-argue it at hour 3.

Version 4: our own decisions

TOPIC	THE QUESTION OR IDEA	WHAT WE DECIDED	WHY
Where the prototype is set	Should the prototype be built for our college's city, since the first round is judged there?	Jharkhand stays the main setting. A small Rajkot region is added, with a live switch.	The sponsor is Jharkhand, which is where the relevance marks come from, and SIH needs no rework. The switch gives local judges something familiar and proves the platform scales.
Working without internet	Team idea: when there's no internet but the phone still has signal, use that; and use GPS for extreme cases to send critical problems	Adopted as the connectivity ladder: coded SMS with exact GPS for critical reports, plain SMS from basic phones, and the offline queue for everything else	Floods usually knock out mobile data first, while SMS usually survives. It also brings in villagers with basic phones.

TOPIC	THE QUESTION OR IDEA	WHAT WE DECIDED	WHY
GPS	Use GPS to send critical problems	GPS can't transmit, so it's used to put an exact location inside the SMS	GPS and NavIC only receive. This keeps the idea honest in front of judges.
Telegram bot	Planned as a Wow feature	Replaced by the SMS channel. Telegram and WhatsApp move to the roadmap.	About the same effort, but SMS works on every phone without data. The SMS call-up still makes a phone buzz on stage.
Tower location and automatic SMS	Use the phone's tower connection	Roadmap, for a native Android app	Browsers can't read tower information or send an SMS without the user tapping Send.
Chatbot	Will the product have a chatbot that guides people and helps with solutions?	No general chatbot. The full product gets three focused assistants; the prototype gets only the solution co-pilot, as the last Wow item.	AI-invented advice in a disaster is dangerous. AI that does specific jobs is more credible with judges.
Team briefing	Every member must understand the whole project	A new "Start here" section at the top of this playbook	Any judge can question any member.

Version 3: the AegisGrid draft

TOPIC	AEGISGRID SAID	WHAT WE DECIDED	WHY
Name and tagline	AegisGrid, "From disaster signal to deployable solution"	JharSetu stays. Its closing image of a bridge is adapted, since that's what <i>setu</i> means.	Same reasons as for ResQGrid: sponsor fit, and a scope wider than disasters. The bridge line makes our name do extra work.
Framing	A disaster-management innovation ecosystem	The innovation lifecycle is adopted (proposals, pilots, adoption). The scope stays broader than disasters.	Its own Q&A says disaster management is "the first use case", which is our framing too.

TOPIC	AEGISGRID SAID	WHAT WE DECIDED	WHY
Priority formula	30 life-safety, 20 people, 15 time, 15 vulnerability, 10 no alternative support, 10 isolation	Our 8-factor formula stays. Its ideas are absorbed: life-safety defines the top severity score, "no alternative support" is our resource gap, and isolation is a vulnerability indicator.	"Life-safety" doesn't fit education or agriculture challenges. Ours also covers hazard exposure and recurrence.
Lifecycle	Ten stages: verified, prioritised, team formed, proposed, expert reviewed, prototype, field tested, adopted, impact measured	A merged 9-stage lifecycle with TEAM_FORMED, SOLUTION_PROPOSED, PILOT and DEPLOYED, plus a fast path for crisis needs	Proposals and pilots are what "project lifecycle" means for university work. Urgent needs can't wait for proposals.
Solutions and Readiness Score	Solutions with a readiness score across seven factors	Adopted as a Must feature and the fifth formula. The demo compares two proposals, 48 against 84.	It turns idea collection into adoption, and makes a memorable demo beat.
Resource-first matching	Ask "what's already nearby?" before "who can solve this?"	Adopted: a resource registry, an "available nearby" panel and PostGIS radius queries, all part of the Capability Matching pitch idea	Cheap to build, and it makes the platform deployable rather than theoretical.
Community validation	Locals mark reports as still a problem, improved, inaccurate, affecting more people, or unsuitable	Adopted. These signals feed confidence, priority, readiness and closure.	It connects ground reality to every score, instead of relying on top-down review alone.
Simulation Mode	A separate simulated-disaster feature for the demo	Merged into Crisis Mode as the mock-drill trigger. The demo button is "Run mock drill".	It gives the demo button a real purpose, since districts run pre-monsoon drills, instead of being a demo-only hack.
Ethical AI explanation	Show why, data used, uncertainty, verifier, suggested	Adopted in the Security section and on the brief card ("AI is unsure about...")	Cheap, and it earns responsible-technology marks.

TOPIC	AEGISGRID SAID	WHAT WE DECIDED	WHY
	action, plus a standard label		
Main demo scenario	Flood medical access with drone delivery	Not the main scenario; the lightning project stays. Drones appear as a mapping resource in the drill, and a drone medicine corridor becomes a follow-up preparedness challenge.	Drone deliveries need aviation approvals, which invites a feasibility attack. Lightning is the state's top killer, and the draft's own readiness principle favours cheap, deployable fixes.
Stack	React + Vite, Express or NestJS, Socket.IO, PostGIS, JWT	Next.js + Supabase stays; PostGIS is adopted as a Supabase extension	Realtime and Auth with row-level security come built in, which avoids the "overbuilt auth" the draft itself warns against.
Roles	Seven platform roles (separate NGO and volunteer, "authority"); a UI/UX and presentation lead on the team	Six platform roles. On the team, UI now also owns deck visuals, the architecture diagram and screenshots.	Fewer role switches on stage. The split of deck work is a good idea.
Schedule	Get the basic flow working before adding AI; feature freeze at H23	The principle is adopted as the H9 checkpoint (the whole path running on the fallback). The freeze stays at H18. Also adopted: failure rehearsals, edge-case tests, and resting the presenter.	A freeze in the last hour leaves no time to rehearse the final product.
Pitch	A five-minute script	Both 5- and 7-minute cuts. Its opening line replaces the ResQGrid second beat, and its resource-first line goes into the architecture segment.	We don't know the time limit, and its opening line is tighter.

TOPIC	AEGISGRID SAID	WHAT WE DECIDED	WHY
Metrics	Headline: time to solution assignment and share reaching pilot or deployment; also "estimated cost saved"	The two headline KPIs are adopted, plus the number of university–industry collaborations. "Estimated cost saved" is dropped.	We can't measure cost saved honestly, and the draft itself warns against fake metrics.
Design theme	An operations-centre look, warm cream ground, red/amber for urgency, teal for coordination, an editorial display font	The direction and colour meanings are adopted. The ground is now neutral, text colours are darker, colour is never used alone, and the font is Mukta.	The app must render Hindi, and needs to pass contrast and colour-blindness checks.
Sustainability	Government SaaS subscription first	Public good first in Jharkhand; white-label licences for other states and NGOs; CSR-funded and sponsored challenges; grants	A government sponsor wants to own the platform, not rent it.
Citizen alerts, and voting on problems	Citizens receive local alerts, and vote on problems	Alerts are roadmap only (relaying official alerts). Voting is replaced by structured community validation, capped at 5% of priority.	Issuing alerts is an official function. Upvotes favour loud areas.
A fix of our own	—	The pilot plan names Jharkhand universities, not our own Gujarat university	A Jharkhand pilot needs local partners. Version 2's wording said "our own university".

Version 2: the ResQGrid draft

TOPIC	RESQGRID SAID	WHAT WE DECIDED	WHY
Name and tagline	ResQGrid, "From Local Need to Collective Action"	JharSetu stays. The draft's tagline becomes the README subtitle, and "See a need. Form a team. Close the loop." becomes the landing headline.	The sponsor is Jharkhand. "ResQ" signals rescue, which invites overlap questions about 112

TOPIC	RESQGRID SAID	WHAT WE DECIDED	WHY
			and the state relief portal, and the problem statement covers far more than rescue.
Framing	A disaster coordination platform	A societal challenge exchange with disaster resilience as the flagship. The draft's disaster coordination becomes Crisis Mode.	The sponsor's background names education, health and agriculture, and asks for routing to universities. A disaster-only product risks "you narrowed the problem".
Main demo scenario	Flood drinking water delivered within 12 hours	The main loop is the Gumla lightning project. The draft's flood moves to Sahebganj as the Crisis Mode finale.	The official list asks for team formation and project lifecycle, which suit a project that runs for weeks. It also keeps one full scenario, which is the draft's own rule.
University role in a crisis	A civil engineering team handles water logistics and distribution	An environmental lab does water testing and planning. The NGO distributes, and industry supplies the units.	Universities don't deliver relief in 12 hours. Judges will spot it.
Stakeholders	Added NGOs, field volunteers and a command-center coordinator	Adopted. There are now six roles, "moderator" is renamed "coordinator", and a state-official persona is added for the district analytics requirement.	Field verification makes the confidence ladder real, and the analytics dashboard is an official component.
Priority formula	35 severity, 25 urgency, 20 people, 10	One 8-factor formula. It keeps the draft's urgency and resource-gap factors and its explicit vulnerability	Equity for remote tribal areas. The resource-gap factor

TOPIC	RESQGRID SAID	WHAT WE DECIDED	WHY
	vulnerability, 10 resource gap	indicators, and our hazard exposure and recurrence. Vulnerability goes up to 15, and popularity is cut to 5.	stops already-covered needs from hogging the top of the queue.
Trust	Confidence ladder, and "never call it a truth score"	Adopted in place of our "trust score". Confidence is kept separate from priority.	"Is it real?" and "how urgent is it?" are different questions, and AI can't answer the first on its own.
Matching formula	40 capability, 25 availability, 15 proximity, 10 resource fit, 10 response time	One formula (40/20/15/15) plus a 10-point partner-type factor: CSR focus and district for companies, track record for universities, response time for NGOs.	One formula is easier to explain on stage, and it still uses CSR fit, which is what brings industry in.
Lifecycle	Nine statuses, including evidence submitted and needs follow-up	Evidence-based closure, NEEDS_FOLLOW_UP and a human-only CLOSED_NOT_ACTIONABLE; REFINED before VERIFIED. <i>Revised again in version 3</i> to add proposals and pilots.	AI drafts first and a human approves second. The draft's rule that a report is never silently rejected is kept.
Report form	Asked for disaster type, available resources and required resources	The form keeps people affected, urgency and vulnerability. The Compiler infers category and hazard, and the coordinator sets resource needs.	The draft's own principle: citizens shouldn't need disaster-management vocabulary, and the form must take under a minute.
Schedule	Deploy by H19, rehearse from H19, feature freeze at H22	Freeze at H18. Deploy the skeleton at H6 with automatic deploys on every merge, and the final deploy by H20. Added from the draft: the H1–H2 design contract, the clickable checkpoint, the clean-browser run at H12, the QA list and the submission checklist.	Freezing after rehearsal means rehearsing a different product from the one you present. Deploying early removes the last-hour surprise.
Team	Five roles,	Six roles, with that owner folded into	SIH teams have six

TOPIC	RESQGRID SAID	WHAT WE DECIDED	WHY
	including a demo, data and QA owner	INT. A five-person fallback is listed.	members.
Stack	React + TS, Supabase or Express, LLM with a deterministic fallback	Next.js (React) + Supabase, plus the draft's "AI suggests, code decides" split and fallback list	Same principles; one repo for UI and API is faster.
Schema and API	14 tables and a REST route list	Merged. Added capabilities, pledges, verifications, evidence, impact records and notifications. Status updates and the audit log are folded into the hash-chained ledger.	One ledger gives us the timeline, the audit trail and tamper evidence in one table.
What not to build	No multilingual voice assistant, no blockchain, no second scenario	Adopted, with clarifications. Hindi voice <i>reporting</i> stays, since it's speech-to-text, not an assistant. The hash chain isn't a blockchain. The crisis finale is a short glimpse.	Keeps our strongest demo moments without breaking the draft's rules.
Pitch	Three repeated ideas; the "absence of coordination" hook; the "complaint vs solution" close	The three ideas map onto our Refine, Match and Deliver pillars, and the "complaint vs solution" line ends the demo. <i>In version 3</i> , AegisGrid's opening line replaced this hook as our second beat.	The Jharkhand-specific opening lands harder with this sponsor.

Sources

- [SIH26043 problem statement summary](#) and the [SIH 2026 problem statement catalogue](#). Also check the official SIH portal.
- [PM's Ten-Point Agenda on Disaster Risk Reduction \(DST\)](#)
- [ETV Bharat: lightning deaths in Jharkhand](#) and [Down To Earth: lightning and tribal communities](#)
- [Aryan Age: the Jharkhand Disaster Relief Management portal](#)